Investify

Project Team

Ahmed Javaid 22I-2530
Rafay Ul Haq 22I-2520
Soban Ahmad 22I-2460

Session _____

Supervised by

Mr. Irfan Ullah



FAST School of Computing

National University of Computer and Emerging Sciences
Islamabad, Pakistan

Submission date: _____

Students' Submission

We submit this report for the FYP-1 Mid-Evaluation of *Investify*, Fall 2026, Department of Software Engineering, FAST–NUCES Islamabad. It documents our implemented modules, evaluation results and remaining development plan. We accept responsibility for the submission and its stated contributions.

Ahmed Javaid (22I-2530) Signature: _____

Rafay Ul Haq (22I-2520) Signature: _____

Soban Ahmad (22I-2460) Signature: _____

Date: _____

Anti-Plagiarism Statement

We declare that sources, reusable software, pretrained models and external data used in this project are acknowledged and referenced. The report distinguishes our implementation and integration work from external components. Development assistance is disclosed in the tool-use statement in Appendix D.5. We accept responsibility for accurate attribution and for the material submitted.

Ahmed Javaid (22I-2530) Signature: _____

Rafay Ul Haq (22I-2520) Signature: _____

Soban Ahmad (22I-2460) Signature: _____

Date: _____

Authors' Declaration

We declare that this report describes the work delivered by the Investify project team for FYP-1 Mid-Evaluation. The member responsibilities and delivered modules are recorded in Chapter 8. Scientific results, limitations and external contributions are identified accurately. Each member accepts responsibility for understanding and defending their contribution and the integrated system.

Ahmed Javaid (22I-2530) Signature: _____

Rafay Ul Haq (22I-2520) Signature: _____

Soban Ahmad (22I-2460) Signature: _____

Date: _____

Abstract

Investify is an educational equity decision-support prototype combining technical indicators, financial-news sentiment and explanations checked against model evidence. Its historical pipeline uses exchange-session alignment, causal features and purged chronological evaluation. A verifier checks displayed claims independently of structured payloads and supplies a checked fallback.

An NVDA experiment covers 1,509 price sessions from January 2018 to December 2023 and 6,768 news rows. Clean five-session macro-F1 is 0.2843 versus a random baseline of 0.3287; a deliberate future-looking feature triggers the leakage control. Independent isotonic calibration produces only Buy decisions. A multiclass temperature diagnostic preserves the raw Buy/Hold/Sell distribution but exposes near-uniform probabilities, so low-confidence cases abstain. Original social classifiers are compared on 24 held-out reference-labelled posts. Risk profiling implements thirteen Grable–Lytton items with capacity constraints and a direction-preserving confidence policy. The live interface demonstrates evidence inspection, profile calculation, explanation rejection and recovery. These results establish implemented controls and traceability; useful prediction, student-model adaptation and participant acceptance remain later evaluation goals.

Acknowledgments

We acknowledge Mr. Irfan Ullah's role as project supervisor and the Department of Software Engineering for the FYP assessment guidance and report template. We also acknowledge the authors of the cited datasets, models and software libraries that support the implementation. Their contributions are identified in the report and references.

Executive Summary

Investify is an educational stock research prototype. It addresses historical information leakage and explanations that misstate a model's evidence. The delivered flow loads a saved case, exposes all technical/news values and SHAP, scores a complete risk profile and checks visible text independently of its structured claims. A rejected explanation retains its error report and produces a separately checked deterministic fallback.

The historical study covers NVDA during 2018–2023. FNSPID remains an earlier comparator; timestamp precision and coverage gaps led to a new Alpaca/Benzinga study. The clean five-session walk-forward macro-F1 is 0.2843, below the stratified-random 0.3287 comparator, and close to the linear 0.2796 result. The centred Bollinger canary detects deliberate future information. Six-fold averages and the final 186-case evaluation are reported separately.

Isotonic calibration improves Brier while emitting all Buy. The separately identified temperature fit reaches T = 1000, giving probabilities near one-third and Brier about 0.6667. This is a no-signal diagnosis. Even the least restrictive real profile threshold is 50%, so all these cases remain Observe. Controlled 0.70-confidence fixtures demonstrate policy behavior without manufacturing a real prediction.

The report answers PF-01–PF-09 through contributions C1–C3, a defined future LLM layer, pinned models, evaluation targets, market mechanisms/formulas, transparent requirements and the live corruption/recovery PoC. The current explainer is deterministic; prompted explanations with constrained claims are scheduled for Mid-2, and QLoRA for Iteration 4.

Ahmed delivered temporal/harness and CI work, Rafay delivered interface/profile and sentiment work, and Soban delivered engine/SHAP and verification work. The equal talk allocation is 220 seconds each. The remaining plan covers broader data/divergence, useful prediction/calibration, service/NFRs and participant evaluation. Existing scientific outputs and receipts retain their identities; new checks and document copies have separate evidence.

Contents

Students' Submission i

Anti-Plagiarism Statement ii

Authors' Declaration iii

Abstract iv

Acknowledgments v

Executive Summary vi

List of Abbreviations xiii

1 Introduction and Project Vision 1

1.1 Project Overview . . . 1
1.2 Problem Statement . . . 1
1.3 Problem Context and Motivation . . . 2
1.4 Project Vision . . . 2
1.5 Project Objectives . . . 2
1.6 Project Scope . . . 3
1.7 Project Modules . . . 3
1.8 Stakeholders and User Classes . . . 4
1.9 Success Criteria . . . 4
1.10 Constraints and Assumptions . . . 4
1.11 Ethical, Legal and DEI Considerations . . . 4

2 Research and Existing Solutions 5

2.1 Research Method / Approach . . . 5
2.2 Existing Solutions / Products . . . 5
2.3 Comparison of Existing Solutions . . . 6
2.4 Limitations and Gaps . . . 7
2.5 Proposed Solution . . . 7
2.5.1 Contribution and Evaluation . . . 7
2.6 Complex Computing Problem . . . 8
2.7 Technical Challenges . . . 8
2.8 Research Findings and Design Decisions . . . 9
Stock Market Background and Design Implications . . . 9
Model Inventory and Adaptation . . . 12

3 Proposal Feedback and Iteration Plan 14

3.1 Proposal Feedback Summary . . . 14

3.2 Proposal Action Register . . . 14
3.3 Iteration Strategy . . . 15
3.4 Iteration 1 Plan . . . 15
3.5 Iteration 2 Plan . . . 16
3.6 Work Breakdown and Responsibilities . . . 16
3.7 Progress Against Plan . . . 16
3.8 Delays, Deviations and Recovery Plan . . . 18

4 Software Requirements **19**
4.1 User Interactions / Use Cases . . . 19
4.2 Use Case Diagram . . . 19
4.3 Detailed Use Cases . . . 19
4.3.1 UC-01: Select a Historical Decision . . . 19
4.3.2 UC-02: Inspect Technical and News Evidence . . . 20
4.3.3 UC-03: Complete the Risk Profile . . . 21
4.3.4 UC-04: Apply the Confidence Policy . . . 21
4.3.5 UC-05: Check an Explanation and Recover . . . 21
4.3.6 UC-06: Inspect a Recorded Study . . . 22
4.3.7 UC-07: Install and Run the Demo . . . 22
4.4 Functional Requirements . . . 23
4.4.1 Data, Decision and Evidence . . . 23
4.4.2 Risk Profiling . . . 23
4.4.3 Explanation and Recovery . . . 24
4.4.4 Research Studies . . . 24
4.4.5 Transparency . . . 25
4.5 Non-Functional Requirements . . . 25
4.5.1 Reliability . . . 25
4.5.2 Usability . . . 25
4.5.3 Performance . . . 26
4.5.4 Security . . . 26
4.6 Acceptance Criteria . . . 27
4.7 Requirements Traceability . . . 27

5 System Design **28**
5.1 System Architecture . . . 28
5.2 Architectural Components . . . 29
5.3 Interfaces and Dependencies . . . 29
5.4 Architecture Decisions and Rationale . . . 30
5.4.1 Explanation Layer (LLM) Design . . . 30
5.4.2 Illustrative Design Example, Not a System Output . . . 32
5.5 Design Evolution . . . 32
5.6 Domain Model . . . 33
5.7 Data Design . . . 34
5.8 Process / Behavior Design . . . 35

6 Iteration 1: Design and Implementation **37**
6.1 Iteration Objectives . . . 37
6.2 Selected Use Cases / Features . . . 37

6.3 Design . . . 37
6.4 Implementation . . . 37
6.4.1 Data Interval and Source Decision . . . 37
6.4.2 Temporal Controls . . . 38
6.5 Completed Features / Module . . . 39
6.5.1 Teacher Construction Pilot . . . 39
6.6 Technical Implementation Details . . . 39
6.7 Algorithms and Technical Decisions . . . 40
6.7.1 Baseline and Leakage Results . . . 40
6.7.2 Calibration Diagnosis and Abstention . . . 42
6.7.3 Complete Risk Profiling and Policy . . . 42
6.7.4 Proposal Gate Outcomes . . . 43
6.8 Tools, Frameworks and Technologies . . . 43
6.9 Demonstration Evidence . . . 45
6.9.1 Current Interface and Actual Captures . . . 45
6.10 Iteration Outcome . . . 49

7 Testing and Early Validation **50**
7.1 Testing Strategy . . . 50
7.2 Test Levels and Techniques . . . 50
7.3 Test Cases . . . 50
7.3.1 Evaluation Metrics and Targets . . . 50
7.4 Test Results . . . 52
7.4.1 Social Results and Reference Construction . . . 52
7.5 Acceptance Testing / Validation . . . 53
7.6 Edge and Failure Cases . . . 53
7.7 Defects and Corrective Actions . . . 53
7.8 Current Validation Status . . . 53

8 Project Management and Development Evidence **54**
8.1 Work Completed . . . 54
8.2 Work Remaining . . . 54
8.3 Individual Responsibilities . . . 54
8.4 Contribution of Each Team Member . . . 55
8.5 Repository and Version Control Evidence . . . 55
8.6 Milestones and Commit Evidence . . . 55
8.7 Tools and Tool-Assisted Development . . . 56
8.8 External Components and Dependencies . . . 56
8.9 Risks and Mitigation . . . 56
8.10 Security and Privacy Considerations . . . 57
8.11 Current Limitations . . . 58

9 Remaining FYP Plan **59**
9.1 Remaining Scope . . . 59
9.2 Remaining Modules and Features . . . 59
9.3 Remaining Iterations . . . 59
9.4 Implementation Plan . . . 59
9.5 Testing and Validation Plan . . . 60

9.6 Deployment Plan 60
9.7 Tentative Timeline 60
9.8 Expected Final Deliverables 60

10 Conclusion **61**

A Proposal Action Register **62**

B Detailed Requirements Traceability **63**
B.1 Transparency Acceptance Anchors 65
B.2 Proposal FR-1–FR-15 to Mid-1 Mapping 65

C Detailed Test Cases **67**

D Repository and Contribution Evidence **69**
D.1 Repository, Revision and CI Evidence 69
D.2 Demo Setup 69
D.3 Scientific Reproduction 70
D.4 Local CI Commands 70
D.5 Document Control and Disclosures 71
D.6 Build and Delivery 71

E Additional Design Diagrams **72**

F Meeting and Progress Evidence **77**

List of Figures

4.1 Implemented learner/operator use cases. 20

5.1 Architecture with the planned LLM layer and the current template path. 28
5.2 Offline record production and interactive explanation/profile checking. 29
5.3 Implemented components and their calls. 31
5.4 Code-linked classes, operations and multiplicities. 33
5.5 File and session relationships. 34
5.6 Verification and independently checked recovery. 35
5.7 Current localhost deployment and downloadable CI output. 36

6.1 Retained leakage-arm lift and intervals; centred and shuffled faults fire, while other variations do not produce guaranteed uplift. 41
6.2 Risk capacity/tolerance calculation and direction-preserving confidence policy. . 44
6.3 Current onboarding purpose, risks, date/horizon and confidence. 46
6.4 Current decision panel with the full Observe headline and a secondary diagnostic class. 47
6.5 Current cautious profile: tolerance 13/47, allowance zero and 80% required confidence. 48
6.6 Current displayed-probability corruption and checked Observe fallback. 49

E.1 Data flow through case, profile and explanation checks. 72
E.2 Checking activity and failure branches. 73
E.3 Verification state and stale-result invalidation. 74
E.4 Current presentation, application, domain and storage layers. 75
E.5 Planned full-product interfaces; not completed software. 76

List of Tables

1.1 Project modules and responsibilities. 3

2.1 Comparison on the five proposal criteria. 6
2.2 Seoul Accord characteristics applied to the project. 8
2.3 Checkpoints, tasks, choices, adaptation, licenses and status. 12

3.1 Panel feedback, action, status and evidence. 14
3.2 Proposal work division and Mid-1 delivery scope. 16
3.3 Proposal versus delivered Iteration 1 and recorded deviations. 16
3.4 Reference iteration milestones and current work. 18

4.1 Data/decision functional requirements. 23
4.2 Profile and policy functional requirements. 23
4.3 Explanation-verification functional requirements. 24
4.4 Study provenance and comparison requirements. 24
4.5 User transparency requirements and acceptance tests. 25
4.6 Preserved proposal NFR identifiers, targets and environments. 26

5.1 Module inputs, outputs and failure conditions. 29
5.2 Illustrative claim checks; no measured model/verifier output. 32

6.1 Consumed historical data and final evaluation interval. 38
6.2 Key files, classes/functions and their behavior. 39
6.3 Actual five-session walk-forward/control results. 40
6.4 Linear comparison read from the retained decision-record metrics. 41
6.5 Held-out calibration behavior on the same 186 evaluation cases. 42
6.6 Supported deliberate corruption and expected recovery. 45

7.1 Definitions, targets and current measurement scope. 51
7.2 Observed local unittest scopes for this revision. 52

8.1 Delivered work and continuation responsibility. 55
8.2 Prospective project risk register adapted from concept Section 17. 56

B.1 Functional requirement, implementation and regression traceability. 63
B.2 Reference FR acceptance and delivered boundary. 65

C.1 Individual input, expected-result and actual-result test cases. 67

D.1 Document control for the revised submission. 71

F.1 Meeting log awaiting actual records. 77

List of Abbreviations

FYP	Final Year Project
FR / NFR	Functional / non-functional requirement
NVDA	NVIDIA stock ticker
RSI	Relative Strength Index
MACD	Moving Average Convergence/Divergence
OBV	On-Balance Volume
SMA	Simple Moving Average
FNSPID	Financial News and Stock Price Integration Dataset
XGBoost	Extreme Gradient Boosting
SHAP	SHapley Additive exPlanations
PoC	Proof of Concept
SRS	Software Requirements Specification
JSON	JavaScript Object Notation
Brier	Brier probability-error score (a name, not an acronym)
CV	Cross-validation
LLM	Large Language Model
QLoRA	Quantized Low-Rank Adaptation
CI/CD	Continuous Integration / Continuous Delivery
API	Application Programming Interface
DEI	Diversity, Equity and Inclusion

Chapter 1

Introduction and Project Vision

1.1 Project Overview

Investify is an educational stock research prototype. A learner selects a historical company decision, inspects the technical and news inputs, completes a risk profile, and checks whether an explanation agrees with the saved evidence. The current interface demonstrates NVDA at a five-trading-session horizon. Price prediction and sentiment scoring were performed offline; profile scoring, explanation checking and recovery run during the demonstration.

1.2 Problem Statement

A retail investor must interpret price history, chart indicators and a large volume of news and social commentary. These signals differ in reliability and can disagree. Without a stated confidence or a record of the inputs, a weak opinion can look as convincing as a strong one, and the investor cannot check how it was formed.

The harder software problem is that historical evaluation can silently use information unavailable at the decision time. A language model can then make that result sound persuasive while inventing a number, a driver or a causal story. Investify addresses these problems through an explicit timing policy, controlled evaluation and explanations checked against the underlying record. Personal financial capacity and willingness to bear loss are considered separately.

1.3 Problem Context and Motivation

The primary user is a learner who understands basic stock-market terms but needs help interpreting model evidence. The operator needs a reproducible installation and a clear failure state when inputs are unavailable. The supervisor needs a demonstrable core module and traceable results. The proposal panel specifically requested contribution clarity, a defined LLM role and transparency throughout the journey.

1.4 Project Vision

Provide a stock research workflow in which the user can inspect the evidence, see uncertainty, understand the role of the risk profile and detect a false explanation. The final product is intended to combine formal news and informal social signals without hiding timing assumptions or replacing weak evidence with fluent certainty.

1.5 Project Objectives

1. Build an evaluation harness whose causal joins, training-only scaling and label-end purging pass named tests, and whose permanent centred-window canary detects deliberately introduced future information.

2. Compare the learned engine with stratified-random and linear baselines using macro-F1, fold variation and probability calibration.

3. Make all five technical and six news features, their definitions and their selected-class SHAP contributions inspectable.

4. Meet the planned explanation targets: numeric fidelity at least 0.98, directional consistency at least 0.95, claim recall at least 0.80 and unverifiable-claim rate at most 0.15, with zero known contradictory emitted claims.

5. Score all thirteen tolerance items and five capacity answers, apply the more restrictive constraint, and preserve the recorded class and probabilities when the profile changes.

6. Evaluate whether intended users can identify the strongest driver, distinguish confidence from return and recognize abstention. These participant measurements belong to the remaining plan.

1.6 Project Scope

Included

The delivered slice includes historical NVDA data, news sentiment, the temporal harness, an XGBoost/linear comparison, SHAP, calibration diagnosis, a full profile, a deterministic template and verifier, and a Streamlit demonstration. A separate MSFT social study compares the original CryptoBERT and Cardiff checkpoints.

Excluded

Brokerage connections, financial transactions and order execution are excluded. Multi-company coverage, integrated social divergence, a prompted LLM explainer, QLoRA, production hosting and participant evaluation are remaining work. The seven equities named in the approved proposal slides remain the intended broader universe.

1.7 Project Modules

Table 1.1: Project modules and responsibilities.

Module	Responsibility
Investor profiling	Capacity, tolerance, restrictive reconciliation and confidence threshold
Sentiment analysis	Headline scoring; social model comparison; future news/social divergence
Predictive model	Causal features, chronological evaluation and class probabilities
Explainability	SHAP, all-feature context, maintained definitions and planned LLM explanation
Explanation verifier	Independent text/claim checks, retained rejection and checked fallback
Conversational interface	Current historical Streamlit flow; future service and follow-up interaction

1.8 Stakeholders and User Classes

The learner operates the evidence and profile screens. The operator installs and maintains the application. Group members prepare the demonstration and continue their assigned modules. The supervisor reviews scope, methodology and deliverables; a supervisor is not a separate software permission role in the current viewer.

1.9 Success Criteria

The demonstration must load valid records, refuse invalid records, show all eleven features, reject supported corruption types, verify its fallback independently and clear stale results after changes. Every walkthrough result must display educational status, date/horizon and limitations. The prediction gate requires out-of-sample evidence against both baselines and calibration; passing software tests alone does not close that gate. Table 7.1 states the measurable evaluation targets.

1.10 Constraints and Assumptions

The available study uses daily historical NVDA prices and headline data from 2018–2023. Source event and acquisition times have different meanings. The saved risk policy is a separate confidence gate and does not personalize the fitted classifier. The scientific environment and recorded-viewer environment are installed separately. Proposal GPU targets use a 16 GB-class device; the inspected local GPU has 10,240 MiB and is a different measurement environment.

1.11 Ethical, Legal and DEI Considerations

The viewer explains that it is an educational prototype and executes no trades. Profile answers remain in local session memory. Source terms and model licenses are recorded before redistribution or later adaptation. Clear wording, explicit unanswered fields and labelled fictional examples reduce dependence on financial literacy. Accessibility and participant consent will be evaluated with the future user study. The combined current limitations are stated once in Section 8.11.

Chapter 2

Research and Existing Solutions

2.1 Research Method / Approach

We reviewed the approved proposal slides, panel action register, official mid-report template and rubric. We then read primary papers, original model cards and official product descriptions, and checked the saved scientific results and implementation. Product descriptions establish advertised features; they do not establish an independently audited evaluation protocol. The unsubmitted concept document is used as design reference rather than as an approved scope change.

2.2 Existing Solutions / Products

Wealthfront uses a questionnaire to recommend an allocation[1]. PortfolioPilot offers portfolio analysis and personalized recommendations[2]. Danelfin exposes technical, fundamental and sentiment subscores[3]. Magnifi offers conversational investment research[4]. Bloomberg and LSEG Workspace combine professional market data and news[5, 6]. StockTwits and TradingView provide social discussion or market-charting tools[7, 8]. These products address different user tasks; Investify evaluates an inspectable historical research path rather than trying to replace a terminal or a licensed adviser.

Ruan and Jiang combine FinBERT-enhanced sentiment, XGBoost and SHAP in a published prediction pipeline[9]. This is close technical prior work, so combining these libraries is not itself our contribution. Li et al. identify training-window information leakage in LLM financial agents[10]. Zhang et al.'s 2026 preprint changes one decision-time convention at a time and finds that inflation depends on the fault[11]; it is not peer reviewed. These studies motivate controlled timing comparisons and a declared null-result path.

2.3 Comparison of Existing Solutions

Table 2.1: Comparison on the five proposal criteria.

Solution	Risk profile	Formal + informal sentiment	Explainable predictions	Leakage-safe evaluation	Verified natural language
Wealthfront	Risk quiz	Not specified	Allocation explanation	Not assessed	Not specified
PortfolioPilot	Risk preferences	Not specified as paired features	Portfolio recommendations	Not assessed	Not specified
Danelfin	Low-risk score; personal profile not established	Sentiment subscore; register separation unclear	Feature subscores	Not assessed	Not specified
Magnifi	Personal help	Not specified	Research chat	Not assessed	Not specified
Bloomberg / LSEG	Risk tools	News; other content varies	Analytics and research	Product-dependent; not assessed	Record-level guard not established
StockTwits / TradingView	Not established here	Social discussion / charts and news	Signals and charts	User-script dependent	Not established here
Published FinBERT + XGBoost + SHAP	Not central	News; no paired social experiment identified	SHAP attributions	Study-specific timing	No independent claim guard identified
Manual workflow	User judgement	User-selected sources	User reasoning	User-dependent	No automatic checker
Investify Mid-1	Complete score + gate	News; social evaluated separately	All-feature SHAP context	Tested controls; Gate A partial	Template grammar + checked fallback
Investify planned	Same profile design	Integrated divergence ablation	SHAP + LLM prose	Broader timing/parity validation	Constrained claims, Mid-2

"Not specified" means the inspected public source did not establish that criterion, not that the product lacks it. The manual row describes the workflow targeted by this project, not a measured user-study baseline.

2.4 Limitations and Gaps

The relevant gap is the combination of time-aware evaluation and a checkable explanation. Showing an indicator, a proprietary score or fluent prose does not by itself show that the inputs were available at decision time or that each explanation claim matches the computation. Conversely, checking a claim does not establish a useful predictor. The current project limitations are consolidated in Section 8.11.

2.5 Proposed Solution

The learner receives the model's probabilities, full evidence and a separate risk-policy outcome. A deterministic verifier checks known claims and retains the reason for rejection before emitting a checked replacement. The planned LLM translates the same record into accessible prose. Its design and evaluation are in Section 5.4.1.

2.5.1 Contribution and Evaluation

1. **C1: Leakage-audited point-in-time harness with a permanent canary.** Implemented timing, scaling and label-end checks; evaluated by unit tests and fixed leakage arms. The centred Bollinger fault fires. Broader historical availability and the full proposal Gate A remain partial.
2. **C2: Deterministic claim verifier with retained rejection and checked fallback.** Implemented for the template grammar; evaluated using independent prose/JSON corruptions, omission tests and the live recovery path. Free-prose extraction is a Mid-2 deliverable.
3. **C3: Cross-register divergence features with a pre-registered ablation.** Planned. Evaluate a news-only model against news/social and divergence variants using shared chronological folds and settings, with the protocol fixed before running the comparison.

As stated on proposal slide 8, no novelty is claimed in sentiment scoring, gradient boosting, SHAP-to-text or QLoRA, and no profit claim is made. The contributions concern controls and their evaluation, not the mere use of these components.

2.6 Complex Computing Problem

The requirement is to justify a stock-research outcome using observations that arrive on different clocks, preserve the information available at decision time, reconcile capacity with tolerance and keep every displayed claim consistent with the underlying computation. These requirements conflict when a fluent explanation or a personalized response would otherwise hide weak evidence.

Table 2.2: Seoul Accord characteristics applied to the project.

Characteristic	Project evidence and boundary
Conflicting requirements	Readability versus checkability; capacity versus tolerance; personalization versus privacy
No obvious solution	Timing and vendor revisions require an explicit information-availability model, not just a random train/test split
Depth of knowledge	Label overlap, calibration, attribution and claim verification require different mathematical models
Unfamiliar issues	Retroactive price adjustment, early closes and revised publication times create silent evaluation faults
Beyond routine practice	Team-built fault controls and record-based rejection tests complement libraries
Diverse stakeholders	Learner, operator and supervisor have different criteria, but the current operating roles are modest
Significant consequences	Unfounded certainty may mislead a non-expert interpreting a stock signal
Interdependence	Store, feature construction and evaluation are coupled; much of the pipeline is still sequential
Ill-defined requirements	Whether short-horizon predictive information exists must be discovered rather than assumed

This is a requirement-level argument adapted from concept Section 5. It does not treat every characteristic as equally strong.

2.7 Technical Challenges

The main challenges are timestamp precision, available-label boundaries, calibration collapse, social-domain transfer and consistent text/JSON claims. XGBoost[12] and

SHAP[13, 14] supply algorithms; the project must control how their inputs, outputs and claims are evaluated. Purged evaluation is motivated by overlapping financial labels[15].

2.8 Research Findings and Design Decisions

The FNSPID precision audit led to a separate timestamped Alpaca/Benzinga experiment. Isotonic calibration collapsed the final classes; the subsequent bounded temperature fit produced near-uniform probabilities and motivated abstention. The full profile is used as a confidence gate rather than changing model direction. The social study motivates evaluating equity transfer before adding divergence. The current template verifier precedes any prompted LLM or QLoRA work[16, 17].

Stock Market Background and Design Implications

Price Formation, Liquidity and Trading Sessions

A share represents an ownership interest in a company. Secondary-market prices arise when buyers and sellers transact through trading venues. A bid is an offered buying price and an ask is an offered selling price; their difference is the spread. Market orders seek execution at available prices, while limit orders restrict the acceptable price. A daily close summarizes completed trading and is not a guarantee that a later order can be filled at that price[18, 19, 20]. A model trained on daily bars cannot infer an intraday order book or assume unlimited liquidity.

Trading takes place on a calendar. US equity sessions contain holidays and early closes; five trading sessions can span more than five calendar days[21]. An announcement after the close belongs to a different information set from one received before it. Our labels use observed price-session endpoints. The inherited fixed 16:00 New York news cutoff is recorded, including its early-close limitation.

Design implication: use an exchange calendar, preserve event and acquisition times and join completed news sessions backward to prices. Evaluate execution assumptions separately before any future trading claim. The present UI reports a historical date and horizon and has no order-execution path.

Price Drivers and the Short-Horizon Noise Problem

Earnings and guidance can change expectations of future cash flows; valuation also depends on the return investors demand. Monetary policy and interest rates can affect dis-

count rates, financing costs and alternatives to equities. Sector and index movements, institutional flows and liquidity can move a stock alongside its own news. These are reasons to avoid explaining a price change from a single headline. The project's eleven inputs contain price/volume summaries and headline sentiment; they are not a complete valuation or macroeconomic model.

Market efficiency treats prices as responding to available information[22]. It does not prove that every market is always efficient or that no prediction can work. It does explain why widely observed short-horizon signals can have little remaining advantage once information, costs and timing are handled correctly. Positive historical performance requires a meaningful comparator and an out-of-sample protocol.

Design implication: compare with stratified-random and linear baselines, preserve weak findings and keep SHAP statements about model use separate from economic causation. A probability is a class estimate, not an expected return. The current no-signal result is a reason to abstain rather than to claim market insight.

The Five Technical Inputs

Let P_t be the completed closing price, V_t traded share volume, and $\Delta P_t = P_t - P_{t-1}$. All windows below end at t; no centred window is allowed in the clean features. Indicator guides describe their usual interpretation[23, 24, 25]; the exact implemented formulas are in `features/indicators.py`.

RSI(14) With $g_t = \max(\Delta P_t, 0)$ and $\ell_t = \max(-\Delta P_t, 0)$, the implementation uses simple fourteen-session averages: $RS = \overline{g}_{14}/\overline{\ell}_{14}$ and $RSI = 100 - 100/(1+RS)$. Completed all-gain/all-loss windows yield 100/0; a completed flat window yields 50. Warm-up remains missing.

MACD histogram $M_t = EMA_{12}(P)_t - EMA_{26}(P)_t$ and $H_t = M_t - EMA_9(M)_t$, where EMA uses $\alpha = 2/(n+1)$. This measures components of price momentum; its sign does not prescribe a trade.

Distance from SMA200 $D_t = (P_t - SMA_{200,t})/SMA_{200,t}$. The long window means early rows do not have a complete value.

Bollinger position $B_t = (P_t - L_t)/(U_t - L_t)$, with $L_t = \overline{P}_{20} - 2s_{20}$ and $U_t = \overline{P}_{20} + 2s_{20}$. The sample standard deviation describes trailing variability. Values can lie outside $[0,1]$; a zero-width band is missing.

OBV slope $OBV_t = OBV_{t-1} + \text{sign}(\Delta P_t)V_t$ and $O_t = (OBV_t - OBV_{t-14})/14$. This is a price-associated volume summary, not an observation of investors' intentions.

Design implication: expose the maintained definition and actual recorded value beside its SHAP effect. A value's sign and its attribution sign answer different questions. Direct perturbation checks verify that changing a future price cannot change an earlier clean feature.

The Six News Inputs, Absorption and Manipulation

FinBERT assigns positive, neutral and negative headline probabilities[26, 27]. For headline i, polarity is $q_i = p_i^+ - p_i^-$. Let n_t be the scored-headline count and a_t the contributing session's average polarity. The implemented six news features are:

$$\text{mean_polarity}_t = a_t = \frac{1}{n_t}\sum_{i=1}^{n_t} q_i, \qquad \text{mention_volume}_t = n_t,$$

$$\text{mom_3d}_t = \overline{a}_{3,t} - a_{t-3}, \qquad \text{mom_7d}_t = \overline{a}_{7,t} - a_{t-7},$$

$$\text{volume_z}_t = \frac{n_t - \overline{n}_{20,t}}{s_{n,20,t}}, \qquad \text{dispersion}_t = s_{a,5,t}.$$

Momentum allows shorter initial rolling windows but still needs the shifted prior value. Attention z-score needs at least five observations and a nonzero standard deviation; dispersion needs at least two. Dispersion is variation in session-average tone, not disagreement between individual headlines. Headline count is news attention, not traded share volume.

News can be anticipated, absorbed quickly or overwhelmed by other information. Tone can also reflect rumours, selective reporting or coordinated discussion. Neither a positive headline nor social agreement proves truth or price impact[28]. The paired social study therefore measures sentiment agreement on its frozen sample rather than investment performance. Cross-register divergence is planned and will be tested with shared evaluation settings.

Design implication: store the later creation/update timestamp for the new news snapshot, disclose coverage and missing features, and use a strict prior-session join. The future explainer must describe conflicting model contributions without inventing absent social values or claiming that a headline caused a return.

Volatility Regimes and Corporate Actions

The distribution of returns and the usefulness of a signal may change between quiet and volatile periods. The same indicator value need not imply the same future outcome. Calibration fitted on one period may transfer poorly to a later refitted classifier, so mode and period identities remain visible.

Splits change share count and price units. NVIDIA's four-for-one 2021 split and ten-for-

one 2024 split are documented by the issuer[29, 30]. The new historical prices use a current-vintage SIP split adjustment, including the vendor's action vintage, without cash-dividend adjustment. A discontinuity in raw prices must not be mistaken for an economic return; adjusted close and total return are different conventions.

Design implication: pin adjustment policy and data vintage, preserve issuer-action references and assess regimes and dividend treatment separately. Compare calibration on its declared fit/evaluation periods. The present model's uniform-confidence outcome supports no directional recommendation across either example profile.

Model Inventory and Adaptation

Table 2.3: Checkpoints, tasks, choices, adaptation, licenses and status.

Model/checkpoint	Task, domain and inputs/outputs	Choice, adaptation, license and status
`ProsusAI/finbert`	Financial headlines to positive/neutral/negative probabilities; polarity is positive minus negative	Chosen for financial text. Original weights, no project fine-tune. News inference completed. Current publisher weights card states CC BY-NC-SA 4.0; pinned-source license context retained.
`ElKulako/cryptobert`	Crypto-social text to three-class sentiment; equity transfer evaluated	Original checkpoint, no project adaptation. MIT model card. Executed on frozen 24-post MSFT held-out set; macro-F1 0.2564.
`cardiffnlp/twitter-roberta-base-sentiment-latest`	Twitter text to three-class sentiment	Alternative social domain. No project adaptation. CC BY 4.0 model card. Same 24 posts; macro-F1 0.6016.
XGBoost project model	Eleven price/news features to Buy/Hold/Sell scores and probabilities	Trained by the project; chronological validation. XGBoost software Apache 2.0. Retained frozen/refit model hashes in scientific receipts.
OpenAI GPT-6 assistant teacher	Saved record/context to controlled prose; six pilot examples	Teacher name comes from frozen plan. Exact serving revision unavailable. Proprietary service terms. JSON targets assembled deterministically; student not trained.

Microsoft Phi-3-mini	Planned local record/context-to-prose student	Original publisher family; exact project checkpoint not selected. MIT publisher model card. Prompted baseline Mid-2; QLoRA Iteration 4[16, 17].
Mistral	Planned fallback local explainer	Exact fallback checkpoint unselected; Mistral-7B-Instruct-v0.3 is an Apache 2.0 publisher candidate. No executed fallback study.

The executed revisions are `0bc2ac10dda6929bcb1f77ccc43c7ec5c424f031` (FinBERT), `9e37c910fe87727cb842a9ac55c6388256fe0f15` (CryptoBERT), and `3216a57f2a0d9c45a2e6c20157c20c49fb4bf9c7` (Cardiff). These are read from the research reference and paired study receipt. Model cards describe domain and licenses[31, 32, 33, 34]; licenses are not inferred from the library used to load a model.

Chapter 3

Proposal Feedback and Iteration Plan

3.1 Proposal Feedback Summary

The panel approved the project with minor modifications. PF-01–PF-09 organize the supplied comments into contribution, LLM role, evidence explanation, problem statement, models, metrics, PoC, transparency and market background. The approved slides provide the proposal claims; the unsubmitted concept supplies additional reference design. Status below describes the technical response, not a supervisor closure decision.

3.2 Proposal Action Register

Table 3.1: Panel feedback, action, status and evidence.

No.	Feedback	Action Taken	Status	Evidence
PF-01	Clear contribution	C1 harness, C2 verifier and planned C3 divergence separated from library inputs	Partial	Section 2.5; leakage results in 6.7
PF-02	Explain LLM benefit	Inputs/outputs, permitted claims, worked example and comparison plan specified	Partial	Section 5.4.1; prompted evaluation Mid-2
PF-03	Explain technical and sentiment signals	All eleven values/definitions and contributions; conflicts and missing social data explained	Partial	Section 2.8; `test_mideval_context.py`
PF-04	Refine problem	Retail workflow, confidence, records, leakage and unchecked fluent text described	Closed	Section 1.2

PF-05	Model clarity	Executed revisions, tasks, domains, adaptation and licenses listed	Closed	Table 2.3; frozen study receipts
PF-06	Metrics	Definitions, thresholds and measured/planned populations separated	Partial	Table 7.1; broader evaluation pending
PF-07	Improve PoC	Full profile, independent checks, supported corruptions and checked recovery demonstrated	Closed	Section 6.9; `test_mideval_policy_explanation.py`
PF-08	Transparency	Onboarding, per-step status/date/confidence/limits and Observe headline requirements added	Partial	MID-FR-21–23; presenter tests; user comprehension pending
PF-09	Market understanding	Price formation, drivers, eleven formulas, absorption, regimes and actions connected to design	Closed	Section 2.8

"Closed" indicates that the requested document/software response is present. PF-01–03 remain partial because divergence, prompted free prose and user benefit are still planned. Appendix A identifies the preserved original feedback.

3.3 Iteration Strategy

Each iteration delivers runnable modules and tests with a separately identified result set. Iteration 1 establishes the data/harness/viewer baseline; Iteration 2 extends engine and services; Iteration 3 adds prompted explanations and measured operations; Iteration 4 compares the adapted student. Weak prediction is a reportable outcome under the proposal gates.

3.4 Iteration 1 Plan

The reference plan calls for hardened ingestion, two-calendar mapping, technical indicators, news features, social selection, an end-to-end linear baseline, a green leakage suite with a firing mutation test, a thin one-company Streamlit view and baselined requirements/design. Table 3.3 records the actual outcome of each item.

3.5 Iteration 2 Plan

Extend multi-company data, evaluate XGBoost/calibration/SHAP against the baseline, preregister the divergence ablation, integrate the service and attempt deployment. The engine and deterministic verifier were built early, while usable prediction, integrated divergence and hosting remain open.

3.6 Work Breakdown and Responsibilities

Ahmed owns temporal data, the leakage harness and CI/delivery; Rafay owns the interface, profile and sentiment path; Soban owns the decision engine, calibration, attribution and verifier/student work. The user confirmed equal delivered contributions. Chapter 8 records delivery and continuing responsibility.

Table 3.2: Proposal work division and Mid-1 delivery scope.

Member	Proposal responsibilities	Mid-1 share
Ahmed Javaid	Point-in-time harness, leakage tests, dual timestamps, parity and deployment	One-third; temporal/CI part delivered
Rafay Ul Haq	Risk profiling, sentiment, degraded operation, acceptance and frontend	One-third; interface/profile/study part delivered
Soban Ahmad	Engine, calibration, SHAP, verifier and QLoRA	One-third; model/checking part delivered

Deployment, participant acceptance and QLoRA are later responsibilities; the delivered column identifies the current evaluation scope.

3.7 Progress Against Plan

Table 3.3: Proposal versus delivered Iteration 1 and recorded deviations.

Reference item	Delivered status	Deviation / reason

Hardened ingestion	Timestamp/dedup/store code and versioned inputs	Current-vintage vendor availability is not historically proven
Two-calendar mapping	Equity/continuous calendar code and tests	Scientific execution is NVDA only; no crypto generalization study
Technical indicators	Five causal indicators	Centred variants exist only as labelled faults
News sentiment features	Six FinBERT headline aggregates	FNSPID/yfinance path replaced for new study by Alpaca/Benzinga/SIP because of timestamp gaps and acquisition recovery
Social selection study	Two original classifiers executed on 24 held-out MSFT posts	AI reference labels replace the proposed hand-labelled sample; independent human agreement absent
Linear baseline end to end	Recorded training/CV/final linear results	UI focuses on saved XGBoost cases; linear result is documented comparison
Leakage suite and mutation	Timing tests and centred Bollinger canary	Gate A partial: five-session clean result is below chance, not literal chance-level closure
Thin Streamlit slice	Five-step historical evidence/profile/checking flow	Expanded beyond original baseline verdict/status slice
Requirements/design baseline	Numbered acceptance and design documents	Formal supervisor approval is not recorded
Asset coverage	NVDA model study; separate MSFT social benchmark	Seven equities in approved slides remain future coverage
Calibration	Isotonic failure retained; temperature diagnostic added	Temperature reaches its bound and gives no usable confidence signal
Risk as model inputs	Full capacity/tolerance calculation and post-hoc gate	Risk is not a classifier feature; $\lambda = 2.25$ from concept was dropped because no loss-aversion utility model is implemented
Teacher construction	Six OpenAI GPT-6 assistant prose examples, deterministic JSON targets	Concept open-weight teacher replaced by recorded assistant session; prompted student and adaptation unexecuted

3.8 Delays, Deviations and Recovery Plan

Timestamp precision, acquisition/runtime recovery, weak clean prediction and calibration collapse caused deviations. Recovery keeps old results, creates separate experiments and prevents a misleading recommendation. Gate A and Gate B outcomes are stated in Section 6.7.4; broader modelling should proceed only with the unmet gate conditions addressed and the null result retained.

Table 3.4: Reference iteration milestones and current work.

Relative checkpoint	Planned deliverable	Owner
Mid-1, week 7 / rubric around week 8	Data/harness, thin UI and this report	All three
Final-1, week 12	Engine/service and divergence evaluation	Ahmed, Rafay, Soban
Mid-2, week 18	Prompted explainer, constrained claims, NFRs/parity and acceptance preparation	Soban, Rafay
Final-2, week 24	QLoRA comparison, external installation and participant evaluation	All three

These are proposal-relative milestones, not invented calendar appointments.

Chapter 4

Software Requirements

4.1 User Interactions / Use Cases

The learner selects a historical case, inspects evidence, completes a risk profile and checks an explanation. The operator installs the viewer and inspects input failures. Research views expose source-bound studies. The current workflow is historical and educational; the final conversational service remains later scope.

4.2 Use Case Diagram

Figure 4.1 uses actors outside the system boundary, oval use cases and labelled include/extend relationships. Study preparation is an operator task, not an unimplemented student actor.

4.3 Detailed Use Cases

4.3.1 UC-01: Select a Historical Decision

Actor Learner.

Precondition The viewer runs and a registered experiment exists.

Trigger Select an experiment and session.

Main flow The loader reads the declared artifact, validates schema and unique sessions, binds its digest and displays class/probabilities for the chosen case.

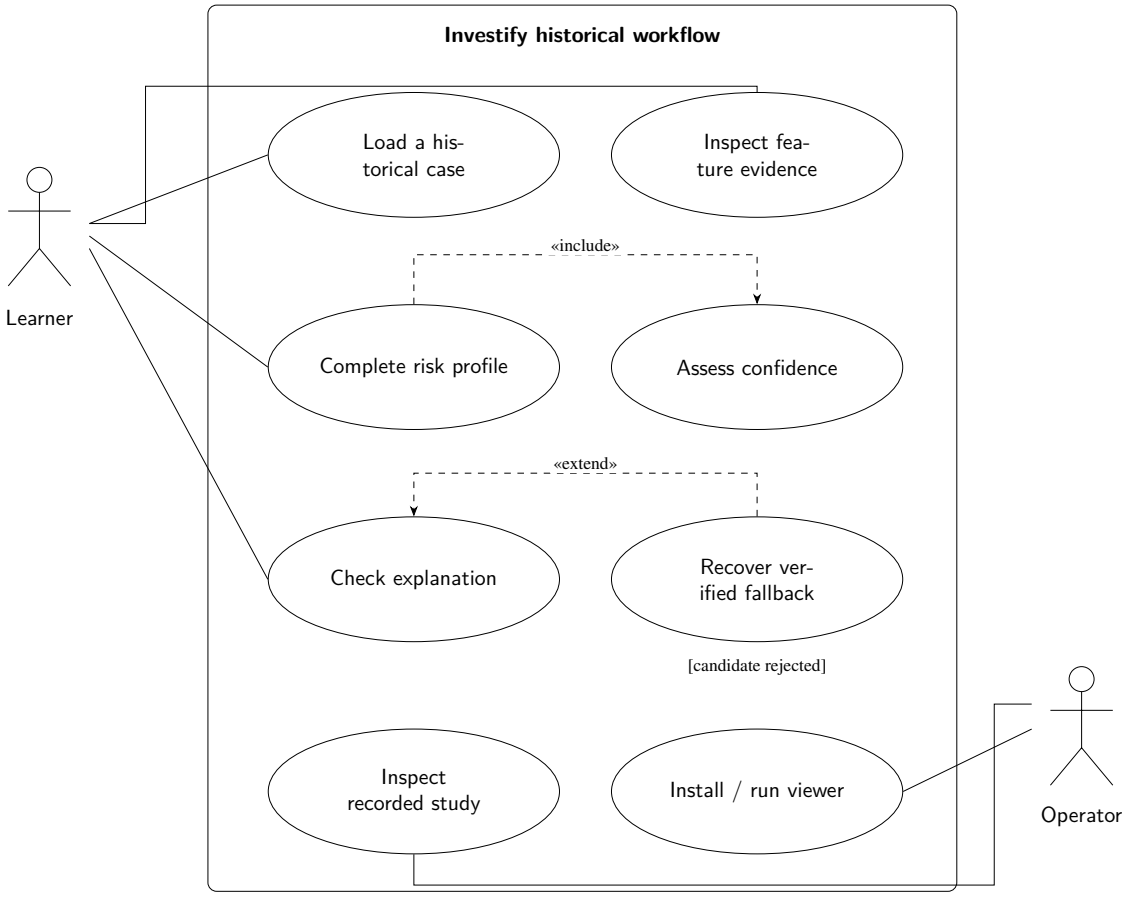


Figure 4.1: Implemented learner/operator use cases.

Exceptions Missing, malformed or changed evidence produces an error without a substituted decision.

Postcondition One source/date/digest identifies the active case; an obsolete explanation result is cleared.

4.3.2 UC-02: Inspect Technical and News Evidence

Actor Learner.

Precondition A valid case is selected.

Trigger Open Evidence and choose a family/feature.

Main flow Display the recorded value, maintained meaning, raw-score contribution and selected-class direction; expose all eleven inputs in detail.

Exceptions A missing value remains explicitly missing; an absent social signal is not fabricated.

Postcondition The learner can trace each displayed feature to the active record.

4.3.3 UC-03: Complete the Risk Profile

Actor Learner.

Precondition Profile form is available.

Trigger Submit five capacity and thirteen tolerance answers.

Main flow Validate all choices, sum published tolerance weights, calculate capacity/caps and reconcile the more restrictive tier. Display the binding axis and allowance R.

Exceptions Any unanswered/invalid choice withholds the result. Explicit fictional examples are labelled; Clear removes the profile.

Postcondition A versioned profile is held only in the local session.

4.3.4 UC-04: Apply the Confidence Policy

Actor Learner.

Precondition Valid case and complete profile.

Trigger Apply/change the profile or selected case.

Main flow Calculate $\tau(R) = 0.80 - 0.30R$ and compare the recorded class confidence. Withhold a directional suggestion below the threshold; never reverse its market direction.

Exceptions Low allowance/confidence produces Observe; an existing Hold signal remains non-directional.

Postcondition The profile outcome is displayed separately from unchanged model probabilities and SHAP.

4.3.5 UC-05: Check an Explanation and Recover

Actor Learner.

Precondition Valid selected record and current candidate text.

Trigger Click Check explanation.

Main flow Check visible and structured claims independently. If accepted, show the current candidate and report. If rejected, retain its failed report, render a deterministic fallback and check it independently.

Exceptions Contradictory, missing, malformed or unsupported claims fail. A fallback/checker failure displays an error, not a successful output.

Postcondition Only a result matching the current case/text is shown; edits invalidate it.

4.3.6 UC-06: Inspect a Recorded Study

Actor Learner or supervisor.

Precondition Retained study receipts and outputs are available.

Trigger Select leakage, sentiment or teacher study.

Main flow Validate identities and coverage, recompute supported summary values and expose sample/scope limits.

Exceptions Changed, incomplete or unavailable evidence withholds replacement scores.

Postcondition Recorded findings are displayed without running model inference.

4.3.7 UC-07: Install and Run the Demo

Actor Operator or group member.

Precondition Git/Python 3.11 and initial package access.

Trigger Follow the setup manual.

Main flow Clone source, create an environment, install pinned viewer dependencies, restore the documented file line endings and start Streamlit on loopback.

Exceptions Missing runtime or occupied port produces an actionable setup error; existing processes are not stopped automatically.

Postcondition The historical workflow runs without model downloads or brokerage credentials.

4.4 Functional Requirements

The IDs preserve traceability while grouping the implementation by responsibility. Profile/calibration revisions are explicitly versioned; the earlier blend remains a historical comparator.

4.4.1 Data, Decision and Evidence

Table 4.1: Data/decision functional requirements.

ID	Requirement	Acceptance
MID-FR-01	Load saved cases; refuse invalid/unavailable records	No synthetic replacement class
MID-FR-02	Expose all technical/news values, meanings and attribution	Five technical and six news inputs inspectable
MID-FR-07	Refuse scientific reruns when structural input/runtime preflight fails	Failure state precedes execution
MID-FR-19	Apply one positive multiclass temperature using earlier outcomes only	Raw argmax preserved; low confidence disclosed

4.4.2 Risk Profiling

Table 4.2: Profile and policy functional requirements.

ID	Requirement	Acceptance
MID-FR-12	Require complete capacity/tolerance answers and expose restrictive reconciliation	No defaults; binding caps/tier displayed
MID-FR-13	Keep historical comparator scaling chronological	Earlier fit only; unsuitable historical sessions refused
MID-FR-14	Keep current risk policy independent of model direction/probabilities	Allowance changes confidence requirement, not Buy/Sell direction

MID-FR-20	Implement all thirteen published tolerance scoring items	Explicit choices; total 13–47; min/max tests pass

4.4.3 Explanation and Recovery

Table 4.3: Explanation-verification functional requirements.

ID	Requirement	Acceptance
MID-FR-03/04	Check prose/JSON independently and reject contradictions/omissions	Correct payload cannot hide wrong prose
MID-FR-05/06	Retain failed report; reject unsupported strict-mode sentences	Candidate/fallback verdicts remain separate
MID-FR-08	Bound explicitly configured generator retries/timeouts	Failure is recorded; fallback labelled
MID-FR-09/11	Check maintained meanings, all-feature values/effects and raw-score scope	Both representations cover required facts
MID-FR-10	Preserve frozen selection and failed/raw attempts	No quality score without actual outputs

4.4.4 Research Studies

Table 4.4: Study provenance and comparison requirements.

ID	Requirement	Acceptance
MID-FR-15	Validate post/reference provenance, exclusions and chronological plan	Declared reference method and fixed identities
MID-FR-16/17	Require complete matching original-classifier outputs and mappings	Partial failures withhold paired metrics

MID-FR-18	Preserve actual teacher prose, independent checking and grouped exports	Test dates excluded; student training separate

4.4.5 Transparency

Table 4.5: User transparency requirements and acceptance tests.

ID	Requirement	Acceptance test
MID-FR-21	Onboarding states purpose, risks and that no trades are executed	Presenter test checks the first stage and the disclosure before Decision
MID-FR-22	Each result step shows advisory status, confidence, historical date/horizon and limits	Presenter test visits Decision, Evidence, Risk profile and Explanation check and checks captions
MID-FR-23	An abstaining case headlines Observe; explanations and fallback preserve the trusted policy outcome	Presenter and policy tests check diagnostic caption, wrong policy prose/JSON and checked recovery

4.5 Non-Functional Requirements

4.5.1 Reliability

Invalid records, missing inputs and failed verification must produce explicit error states. Case, candidate and profile changes invalidate old results. No known contradictory candidate may be emitted as a verified result.

4.5.2 Usability

Each walkthrough step must state what the displayed value means and its limits. Questions begin unanswered, and fictional examples are labelled. Participant acceptance will

measure completion, time and interpretation errors.

4.5.3 Performance

Table 4.6: Preserved proposal NFR identifiers, targets and environments.

ID	Target	Environment/status
NFR-1	Decision, attribution and checking below 800 ms p95	Target VM, warm cache; unmeasured
NFR-2	Sentiment at least 150 documents/s	16 GB-class GPU, fp16, 128 tokens; unmeasured
NFR-3	TreeSHAP below 50 ms per prediction	CPU; target measurement pending
NFR-4	Peak inference memory below 4 GB	Target VM; unmeasured
NFR-5	QLoRA at most four GPU-hours within 16 GB	Iteration 4; local 10,240 MiB GPU is a different budget, not evidence that this target was met
NFR-6	Explanation including verification below five seconds	Target VM CPU; prompted path unimplemented
NFR-7	Nightly full-universe ingest below 30 minutes	GPU plus local store; operational task pending
NFR-8	Purged walk-forward/search below 90 minutes	CPU; full target sweep pending

The scientific `pyproject.toml` uses NumPy 1/pandas 2; the recorded viewer's pinned requirements use NumPy 2/pandas 3. They are separate environments. A short test run is not a benchmark of the specified NFR workload.

4.5.4 Security

Profile answers are local to the session; secrets and model caches are excluded from packaged source. Loopback is the default network binding. Production authentication, threat modelling and source-rights review are remaining service requirements.

4.6 Acceptance Criteria

A correct supported explanation passes; wrong visible prose fails beside correct JSON; a rejected candidate retains its report and produces an independently checked fallback. Unanswered profiles withhold results. Profile changes preserve the model direction/probabilities and invalidate prior explanations. An unavailable study withholds replacement scores.

4.7 Requirements Traceability

Appendix B maps source functions and named tests to the Mid-1 IDs and maps the fifteen concept-reference FRs to the delivered slice. MID-FR-21–23 add observable transparency acceptance. Counts in Chapter 7 come from executed suites rather than from the number of requirement rows.

Chapter 5

System Design

5.1 System Architecture

The product architecture separates data/sentiment, prediction, SHAP/context, explanation, verification, profile policy and the interface. The implemented path uses the deterministic template. The dashed Explanation layer (LLM) is scheduled for Mid-2 between context and the verifier.

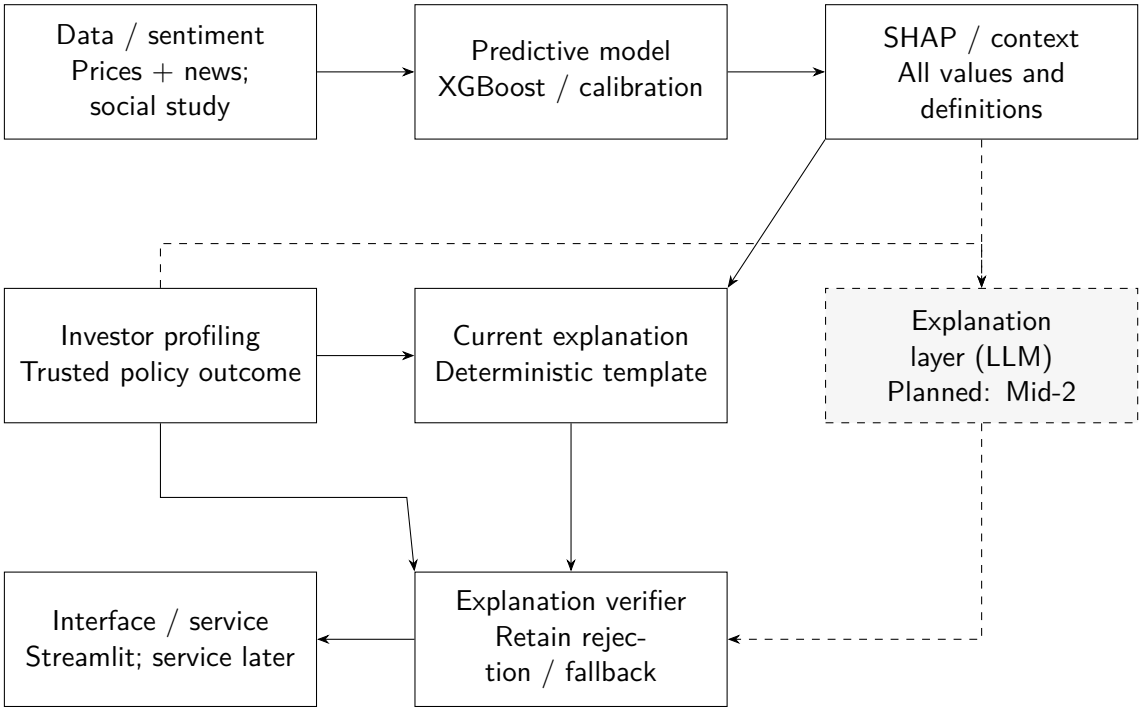


Figure 5.1: Architecture with the planned LLM layer and the current template path.

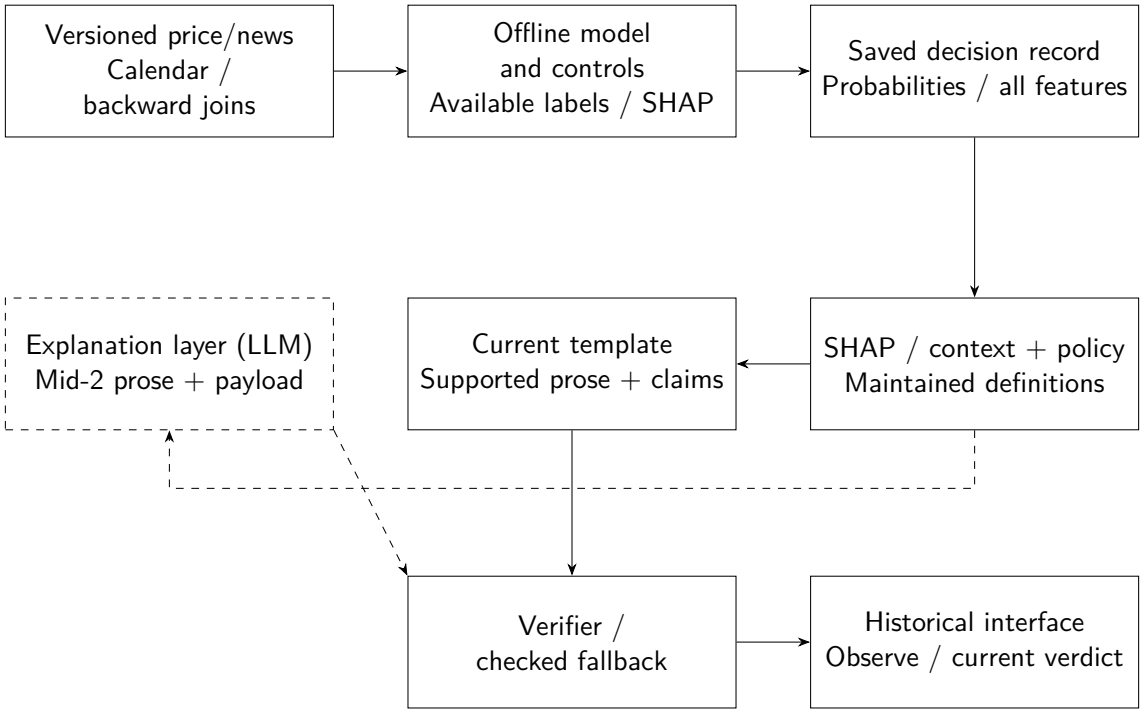


Figure 5.2: Offline record production and interactive explanation/profile checking.

5.2 Architectural Components

The loader validates saved records. Context exposes eleven values, SHAP and maintained definitions. Profile scoring computes allowance; the confidence policy produces an independent outcome. The template and verifier operate on the same record. The UI binds visible results to the current case/profile/text.

5.3 Interfaces and Dependencies

Table 5.1: Module inputs, outputs and failure conditions.

Interface	Input/output	Failure condition
Record loader	JSON to validated DecisionRecord collection	Missing, malformed or inconsistent source
Evidence context	Record to values/definitions/SHAP map	Missing values remain explicit
Risk policy	Record + complete profile to Observe/Buy/Hold/Sell	Ineligible model or inadequate confidence

Template/checker	Record, text/claims, trusted policy to claim verdicts	Wrong, missing or unsupported claims
Emission	Candidate and verification to accepted text or checked fallback	Fallback failure is an error
Planned LLM	Context + policy to prose and constrained payload	Invalid output is rejected; no alternate decision invented

The viewer depends on Streamlit and the core Python modules. Scientific execution separately depends on pandas, XGBoost, native SHAP, calendars and publisher sentiment weights. No model download is required to inspect saved records.

5.4 Architecture Decisions and Rationale

Saved historical outputs keep the demonstration reproducible. A strict prior-session join makes the timing rule inspectable. Label endpoints and calibration availability prevent future outcomes from entering fit stages. The profile is a separate gate so it cannot reverse the recorded market class. The verifier is deterministic so an LLM does not grade its own claims. Rejection and fallback have separate reports so a successful replacement cannot conceal the original error.

5.4.1 Explanation Layer (LLM) Design

Inputs: company/session/horizon, class probabilities, all eleven technical/news values, selected-class SHAP values and ranks, maintained feature definitions, missing-data flags and the trusted risk-policy outcome. Future integrated records may add news/social divergence; absent social evidence must remain absent.

Outputs: plain-language prose plus a structured claims payload with record keys, claim types and the corresponding prose spans. The LLM must explain technical and sentiment signals, opposing contributions, uncertainty and what the data does not show. It cannot choose a new class, change the policy outcome or add a market fact not supported by the record.

Permitted claims: recorded values/probabilities, maintained indicator meanings, SHAP support/opposition for the selected raw class, missing coverage and the policy outcome. **Forbidden claims:** guaranteed returns, causal assertions from SHAP alone, confidence as profit, invented headline/social content, personal suitability or trades being executed.

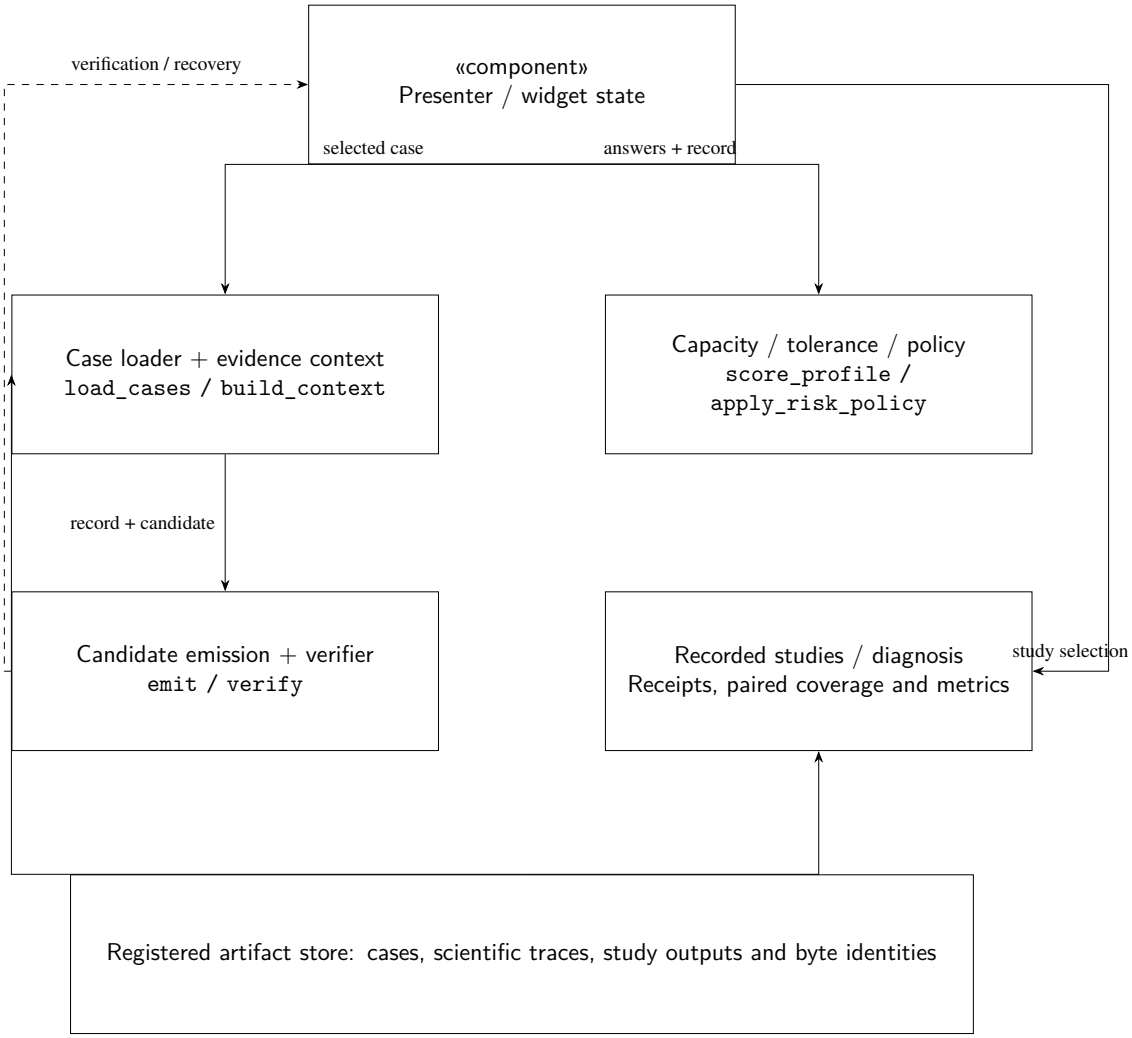


Figure 5.3: Implemented components and their calls.

Verification and fallback: check numbers, directions, ranks, feature names and policy against trusted inputs; reject contradictory or missing essential claims. Bound regeneration attempts and separately verify the deterministic replacement. The current verifier parses only the template sentence grammar. Extracting and binding claims from free prose through a constrained payload is the Mid-2 deliverable, not a current capability.

Planned evaluation: compare deterministic template, raw prompted Phi-3-mini and later adapted student on the same frozen cases. Score raw output before filtering and emitted output after filtering; report the four proposal metrics, contradictions, retries, fallback rate and whole-request latency. Participant tasks test whether prose improves interpretation beyond the template. QLoRA is scheduled for Iteration 4.

5.4.2 Illustrative Design Example, Not a System Output

This adapted concept Section 10.8 example uses hypothetical inputs: Hold probability 0.58, news polarity +0.31, social polarity -0.18 and RSI 38.2. Suppose divergence contributes +0.14 and RSI contributes -0.06 to the raw Hold score. A future explanation could say: "The diagnostic class is Hold with 58% confidence. News and social tone disagree. The recorded divergence supports Hold, while the RSI contribution opposes Hold. These are model effects; they do not establish what caused the price to move." The displayed policy must come from a separately supplied trusted policy result.

Table 5.2: Illustrative claim checks; no measured model/verifier output.

Claim	Planned check	Design judgement
Hold at 58%	Class/probability versus record	Supported if inputs match
News +0.31; social -0.18	Feature values and coverage	Supported only with integrated inputs
Divergence supports Hold	Selected-class SHAP sign +0.14	Supported model-effect wording
RSI opposes Hold	Selected-class SHAP sign -0.06	Does not prove support for Buy
Volatility caused the move	No causal evidence supplied	Refuse unsupported claim

The current template checker does not accept this unrestricted paragraph. No hypothetical pass rate or fidelity score is reported for it.

5.5 Design Evolution

The proposal's broader conversational pipeline remains the destination. The delivered slice adds complete evidence, independent text/JSON checking, a full profile and abstention. Isotonic failures and the earlier risk/sentiment blend are retained as historical comparisons. Generated explanations and divergence require separate integration/evaluation.

5.6 Domain Model

The classes identify record collections, individual feature contributions, explanations and verification results. A profile depends on a record through the policy; it does not mutate classifier probabilities.

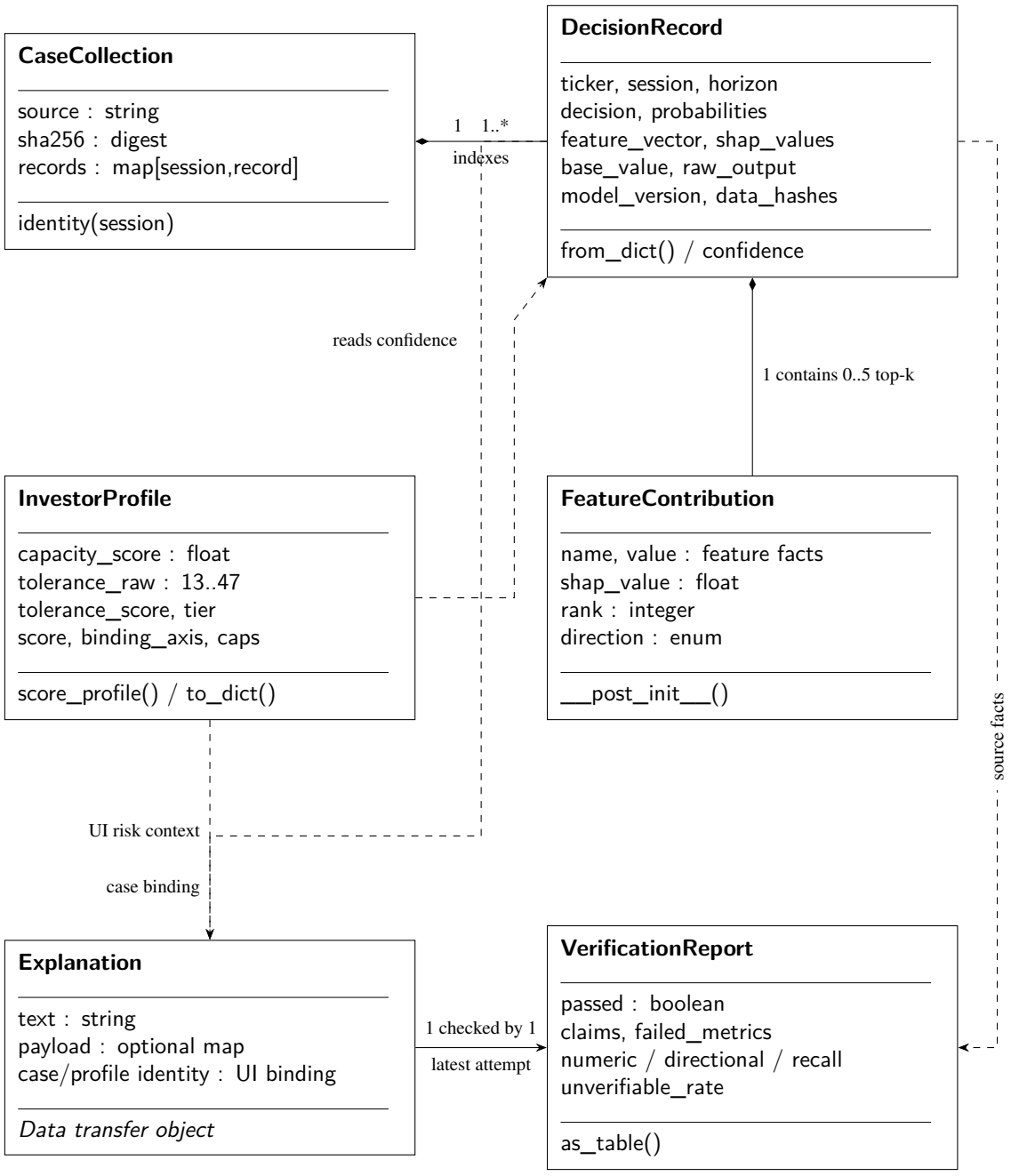


Figure 5.4: Code-linked classes, operations and multiplicities.

5.7 Data Design

Experiment source/digest and session jointly identify a case. Feature maps retain names, values and contributions. Candidate/profile digests bind verification to the current state. JSON files and session memory are the current storage; no relational database is implemented.

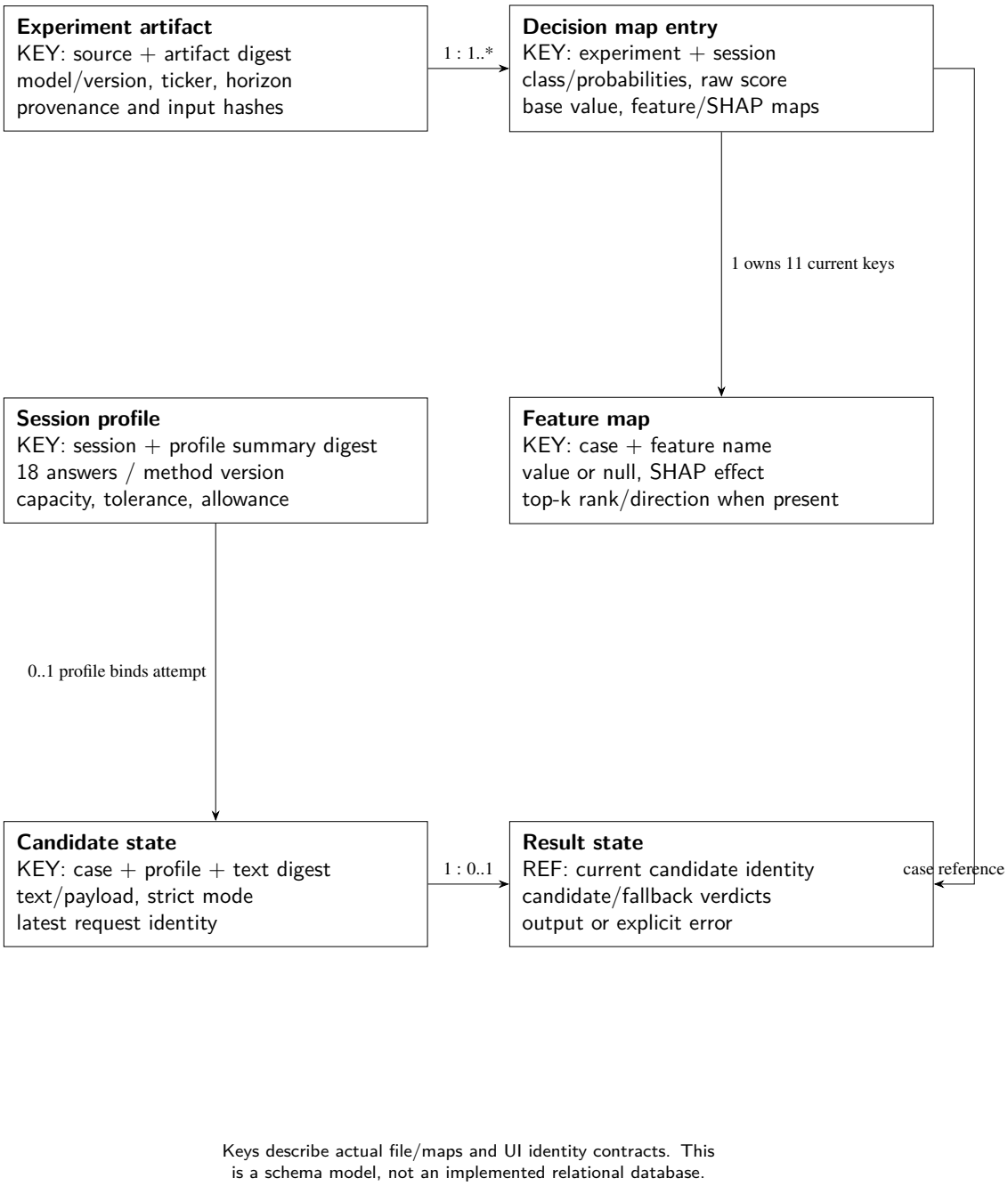


Figure 5.5: File and session relationships.

5.8 Process / Behavior Design

The sequence starts at Check explanation and has acceptance, rejection/fallback and failure branches. The activity and state models also represent invalid input and stale verification. The extra diagrams are collected in Appendix E.

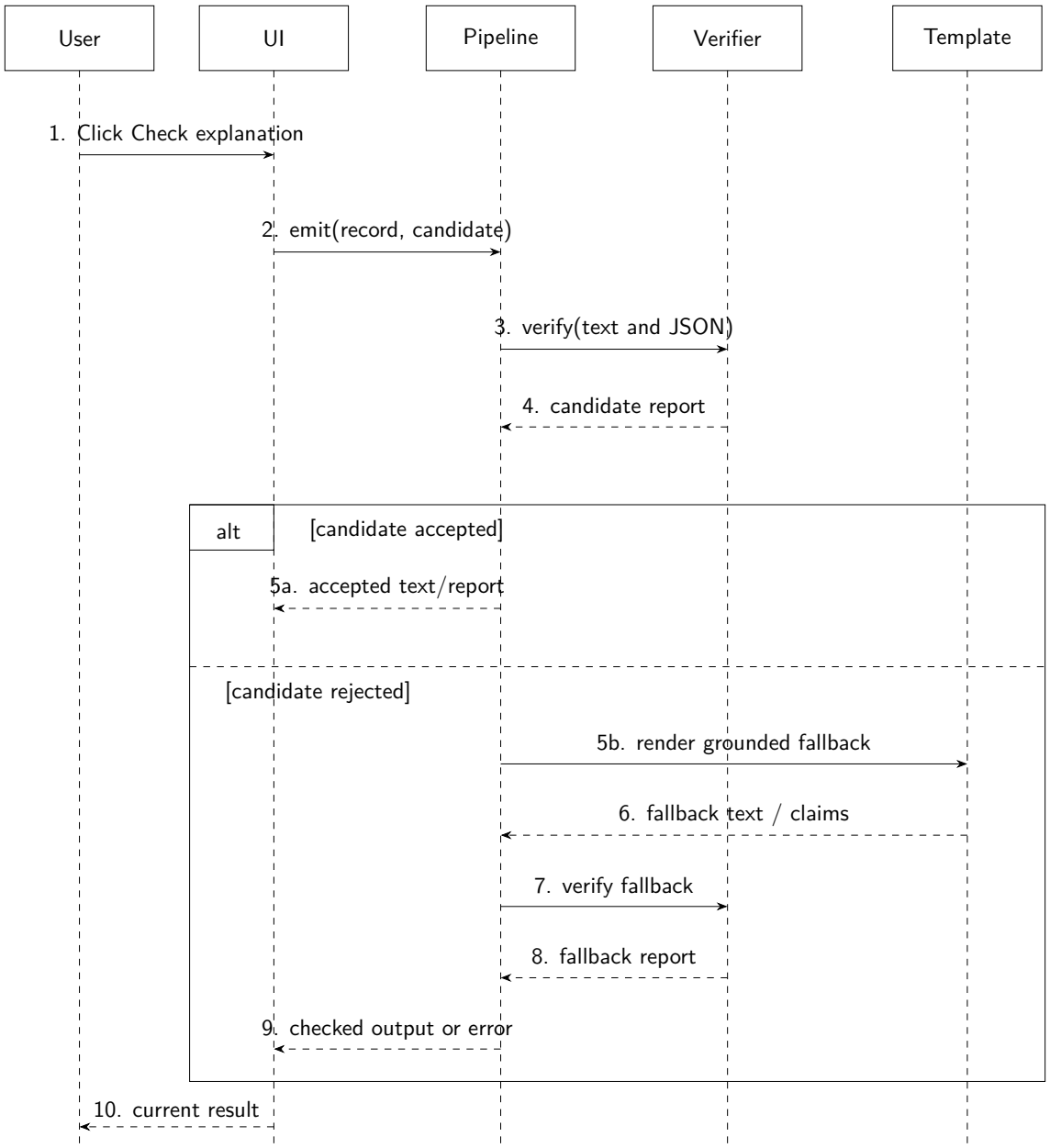


Figure 5.6: Verification and independently checked recovery.

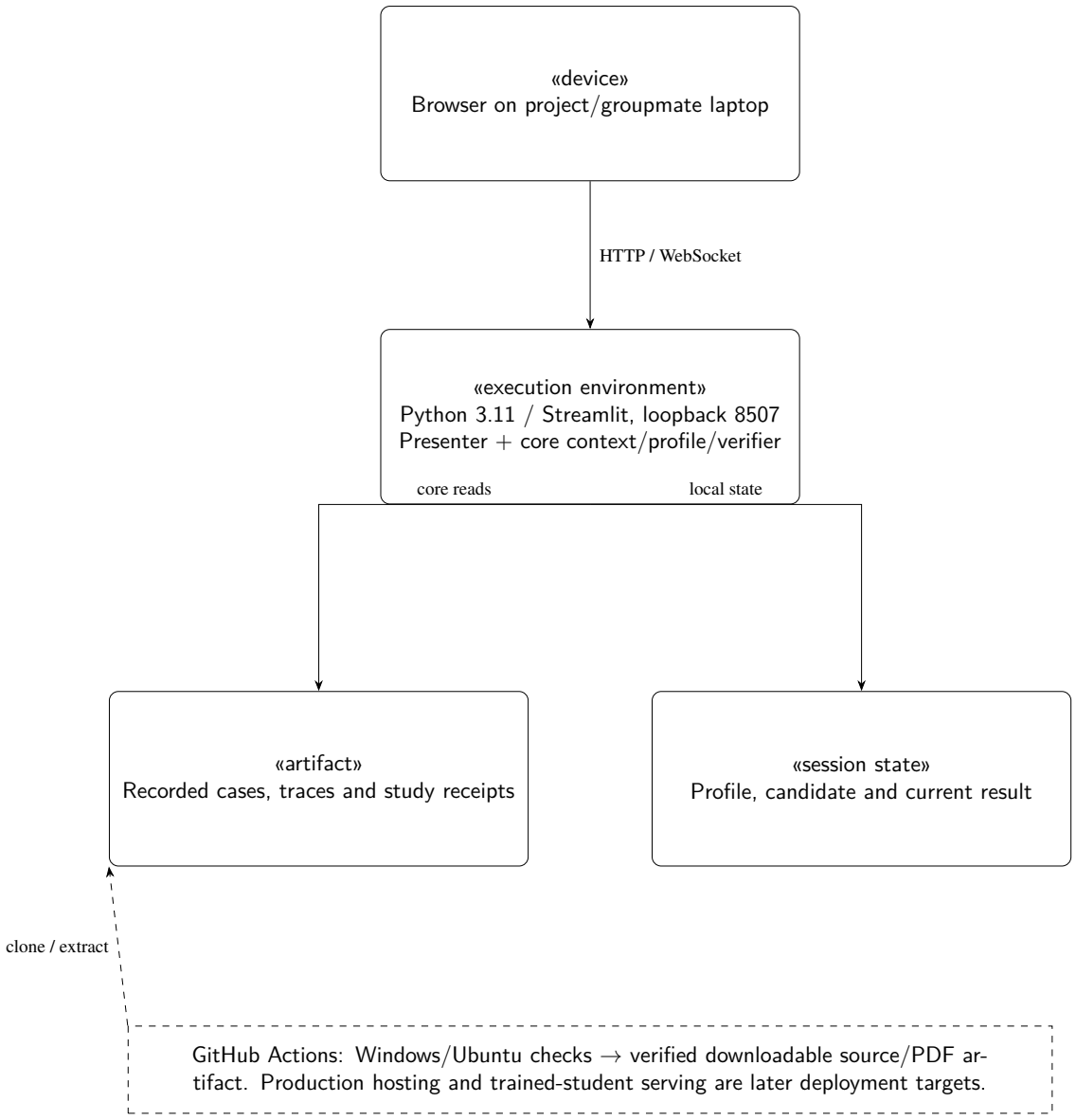


Figure 5.7: Current localhost deployment and downloadable CI output.

Chapter 6

Iteration 1: Design and Implementation

6.1 Iteration Objectives

Deliver a reproducible historical research slice with causal features, a baseline and leakage controls, inspectable attributions, a complete profile and an explanation-checking demonstration. Identify unmet proposal gates before extending the product.

6.2 Selected Use Cases / Features

UC-01–UC-07 cover case selection, evidence, capacity/tolerance scoring, confidence policy, checking/recovery, recorded studies and installation. The main demonstration selects the frozen multiclass NVDA case for 21 December 2023.

6.3 Design

The offline path produces saved decision records. The live viewer uses the same record for evidence, profile policy and explanation checks. Date, profile and candidate identities prevent a result for an earlier state from appearing as current. Chapter 5 shows the planned LLM branch separately.

6.4 Implementation

6.4.1 Data Interval and Source Decision

Table 6.1: Consumed historical data and final evaluation interval.

Input	Interval/count	Policy
NVDA daily prices	2 Jan 2018–29 Dec 2023; 1,509 sessions	SIP split-adjusted snapshot; cash dividends not adjusted
Alpaca/Benzinga news	2018–2023; 6,768 retained rows, 5,748 distinct headlines	Availability is later creation/update time
Model evaluation	29 Mar–21 Dec 2023; 186 decisions per mode	Five observed-session outcomes
Calibration	181 available earlier outcomes	Five outcomes crossing evaluation start excluded

FNSPID remains a preserved historical comparator. Its original audit found that 99.78% of the selected NVDA rows carried midnight-only timestamps, with only 26 genuinely intraday-timed rows; AAPL/TSLA also had material year gaps. Those limitations cannot support a broad precise-publication alignment claim. Alpaca/Benzinga was used for the new timestamped study, consistent with the original repository's documented recovery. The scope remains NVDA until multi-company coverage is audited. Current-vintage data is not claimed as a byte-identical September reproduction.

6.4.2 Temporal Controls

The target spans five observed trading sessions. Purging excludes a training label whose actual endpoint reaches the evaluation start. The configured five-day embargo is an additional calendar-day gap: these are different rules. Replacing endpoint logic with calendar subtraction retained 18 invalid endpoints on the boundary fixture, while the corrected purge retained zero. The explanation of this distinction resolves the proposal/report terminology difference rather than treating the two terms as synonyms.

The no-purge arm retains an additional gap and may show no detectable performance inflation when only a few boundary outcomes are affected. The same-bar branch remains a lag ablation under the strict-prior join, not a guaranteed future leak. A centred window explicitly consumes future observations and is the permanent canary. The shuffled split breaks chronological evaluation. Each arm has a different fault geometry; not every deliberate variation must inflate the measured score.

6.5 Completed Features / Module

The delivered module loads and validates cases, shows eleven inputs and SHAP, scores all profile answers, applies a separate confidence threshold and verifies supported explanations with checked fallback. A deterministic template is implemented; a working LLM explainer is not claimed.

6.5.1 Teacher Construction Pilot

The frozen packet plan names OpenAI GPT-6 assistant as teacher. Six recorded controlled-prose examples are retained: four training and two validation. Factual JSON is assembled from the record rather than generated by the teacher. Grouped dates exclude final test sessions. This is example construction, not prompted-student inference or QLoRA.

6.6 Technical Implementation Details

Table 6.2: Key files, classes/functions and their behavior.

File	Implementation detail
`decision_record.py`	DecisionRecord validates probabilities, class, feature maps and ranked contributions; from_dict is the read boundary
`explain/context.py`	FEATURES supplies eleven maintained definitions; build_context retains missing values and raw-score scope
`explain/template.py`	render/render_guided produce supported text and structured claims, including the trusted policy outcome
`explain/verifier.py`	verify checks displayed claims and payload independently; strict mode rejects unparsed lines
`explain/pipeline.py`	emit retains rejected candidate/report and verifies the deterministic fallback separately
`risk_profile_gl.py`	score_profile validates eighteen answers, applies caps and restrictive allowance; apply_risk_policy preserves class/probabilities

`multiclass_calibration.py`	fit_temperature uses earlier available outcomes; softmax_temperature preserves argmax for positive temperature
`harness/splits.py`	Expanding walk-forward split uses actual label endpoints and a separate calendar-day gap
`harness/arms.py`	Defines clean, shuffled, no-purge, centred and lag variants on shared rows/settings
`features/causality_audit.py`	Perturbs future observations to check that earlier clean features are invariant
`calendars.py`	Maps session-bound and continuous time windows; the executed equity study uses NYSE sessions

The frozen context and scientific sources retain their recorded identities. Policy-aware explanations use a new verifier version; original teacher source snapshots are retained and packet replay is tested.

6.7 Algorithms and Technical Decisions

6.7.1 Baseline and Leakage Results

Table 6.3: Actual five-session walk-forward/control results.

Arm	Macro-F1	Lift over random	Interpretation
Clean	0.2843	-0.0444	Below random comparator
Shuffled folds	0.4457	+0.1142	Chronological fault shows signal
No purge	0.3022	-0.0267	No measured inflation under this geometry
Centred Bollinger	0.5390	+0.2104	Explicit future-information canary
Centred SMA200	0.3333	+0.0046	No detectable uplift for this variation
Same-bar text	0.2651	-0.0636	Lag ablation, not evidence of future access

The centred-Bollinger lift interval is [+0.1698,+0.2421]. Results use fixed settings, shared

row masks and six folds. Weak clean performance is retained; the experiment does not establish a useful forecasting strategy.

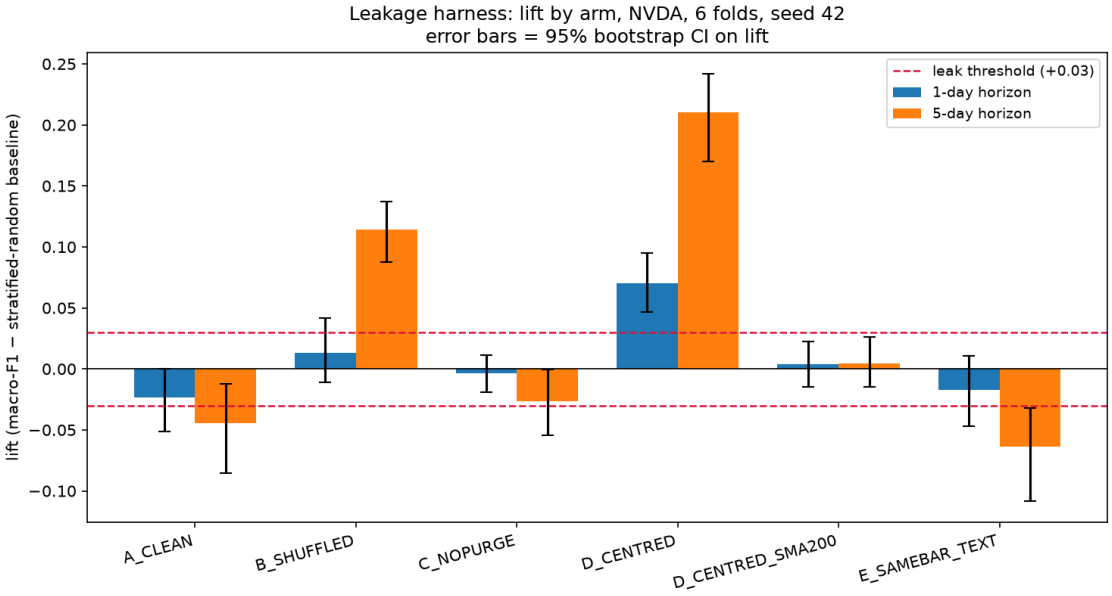


Figure 6.1: Retained leakage-arm lift and intervals; centred and shuffled faults fire, while other variations do not produce guaranteed uplift.

Table 6.4: Linear comparison read from the retained decision-record metrics.

Scope	XGBoost macro-F1	Linear macro-F1
Six-fold walk-forward CV	0.2843 ± 0.0537	0.2796 ± 0.0559
Final 186 cases, frozen isotonic	0.2238	0.1800
Final 186 cases, refit isotonic	0.2238	0.1896

The reported standard deviations are fold standard deviations, not confidence intervals. Both frozen and refit files carry the same training CV summaries; they are not two independent CV studies. The 0.2843 result is the six-fold walk-forward mean, whereas 0.3182 is the raw frozen model's macro-F1 on the separate final 186 evaluation cases.

6.7.2 Calibration Diagnosis and Abstention

Independent isotonic maps force Buy because its pre-normalization minimum 0.5344828 exceeds Hold maximum 0.2500000 and Sell maximum 0.3139535. Normalizing by a common positive sum preserves that ordering. We retain those records as the diagnosed failure and construct a new global-temperature experiment from the same immutable model bytes, scaler, margins and input rows.

Table 6.5: Held-out calibration behavior on the same 186 evaluation cases.

Method	Buy/Hold/Sell	Macro-F1	Brier
Frozen raw	40 / 51 / 95	0.3182	0.7929
Frozen isotonic	186 / 0 / 0	0.2238	0.6105
Frozen temperature	40 / 51 / 95	0.3182	0.6667
Refit raw	37 / 50 / 99	0.3116	0.7760
Refit isotonic	186 / 0 / 0	0.2238	0.6078
Refit temperature	37 / 50 / 99	0.3116	0.6667

The single temperature is selected by negative log likelihood on the 181 earlier calibration outcomes; evaluation labels do not fit it. Its optimum reaches the configured upper bound of 1,000, making probabilities nearly uniform. Class order is preserved, but probabilities approach a constant one-third forecast. Its multiclass Brier score is about 0.6667, the constant $1/3$ forecast value $\sum_{k=1}^{3}(1/3-y_k)^2 = 2/3$. The observed value is slightly higher before rounding. There is no usable confidence signal, so the policy abstains. The interface therefore exposes the diagnostic class and withholds a directional recommendation at insufficient confidence. Brier and classification metrics are reported together; this is a no-signal diagnosis, not a usable forecast.

Native XGBoost replay reproduces the retained margins. TreeSHAP is recalculated for each newly selected raw class and checked for additivity. Both numeric and complete-guided explanations are checked on all newly generated records. The new experiment identity keeps the original classifier and calibration findings traceable.

6.7.3 Complete Risk Profiling and Policy

The tolerance instrument originates in Grable and Lytton[35], with the scoring reviewed through the later retrospective[36]. Five capacity factors use retained weights: horizon/buffer 0.30 each, income stability 0.20, dependants/age 0.10 each. Protective caps handle no buffer, short horizon and unstable income with thin reserves. Thirteen Grable–

Lytton item scores produce $G \in [13,47]$ and normalized tolerance $T_r = (G-13)/34$. The risk allowance is $R = \min(C, T_r)$; the more restrictive capacity/tolerance tier is displayed.

The primary policy requires confidence $\tau(R) = 0.80 - 0.30R$ for a directional suggestion. A higher allowance lowers that threshold but cannot change Buy into Sell or Sell into Buy. Weak evidence remains Observe; an existing Hold class remains non-directional. The old $C = wS + (1-w)T$ blend remains a historical baseline with explicit weighting, not the meaning of the current risk score. The published scoring is implemented; local paraphrases and the confidence policy require further suitability evaluation.

On real temperature records, even maximum allowance gives a 50% threshold against confidence near 33%; the profile never changes their Observe outcome. Controlled 0.70-confidence Buy/Sell fixtures show the intended policy behavior separately: higher allowance can clear the threshold without reversing direction or changing probabilities.

6.7.4 Proposal Gate Outcomes

Gate A: partial, not fully closed. The reference gate requires timing/session checks, fold-local scaling, both negative controls at chance and a firing mutation test. The saved one-session clean arm is PASS (0.3079 versus random 0.3314); the centred canary fires at both horizons. The five-session clean arm is BELOW_CHANCE (0.2843 versus 0.3287), so it is not a literal chance-level pass for the whole gate. Original historical vendor availability is also unproven. Structural timing checks and a firing canary are supported; unconditional leakage-free closure is not.

Gate B: not passed. It requires macro-F1 and calibration improvement over the linear baseline under purged out-of-sample evaluation, with controls still at chance. The CV macro-F1 difference is only 0.0047 and clean prediction is below stratified random. The final isotonic classification lifts do not establish the joint CV calibration criterion; temperature reduces confidence to approximately one-third. The proposal's null-result path applies: retain the harness/verifier contribution and continue analysis without claiming useful prediction.

6.8 Tools, Frameworks and Technologies

Python, pandas/NumPy, XGBoost, native TreeSHAP, exchange calendars, original sentiment classifiers, Streamlit, unittest, Ruff, Git and GitHub Actions support the implementation. LaTeX/Beamer and the unchanged institutional class/logo produce the documents. The prepared compiler uses XeTeX with bibliography processing through Tectonic; xelatex/bibtex command-line executables are not installed on this workstation.

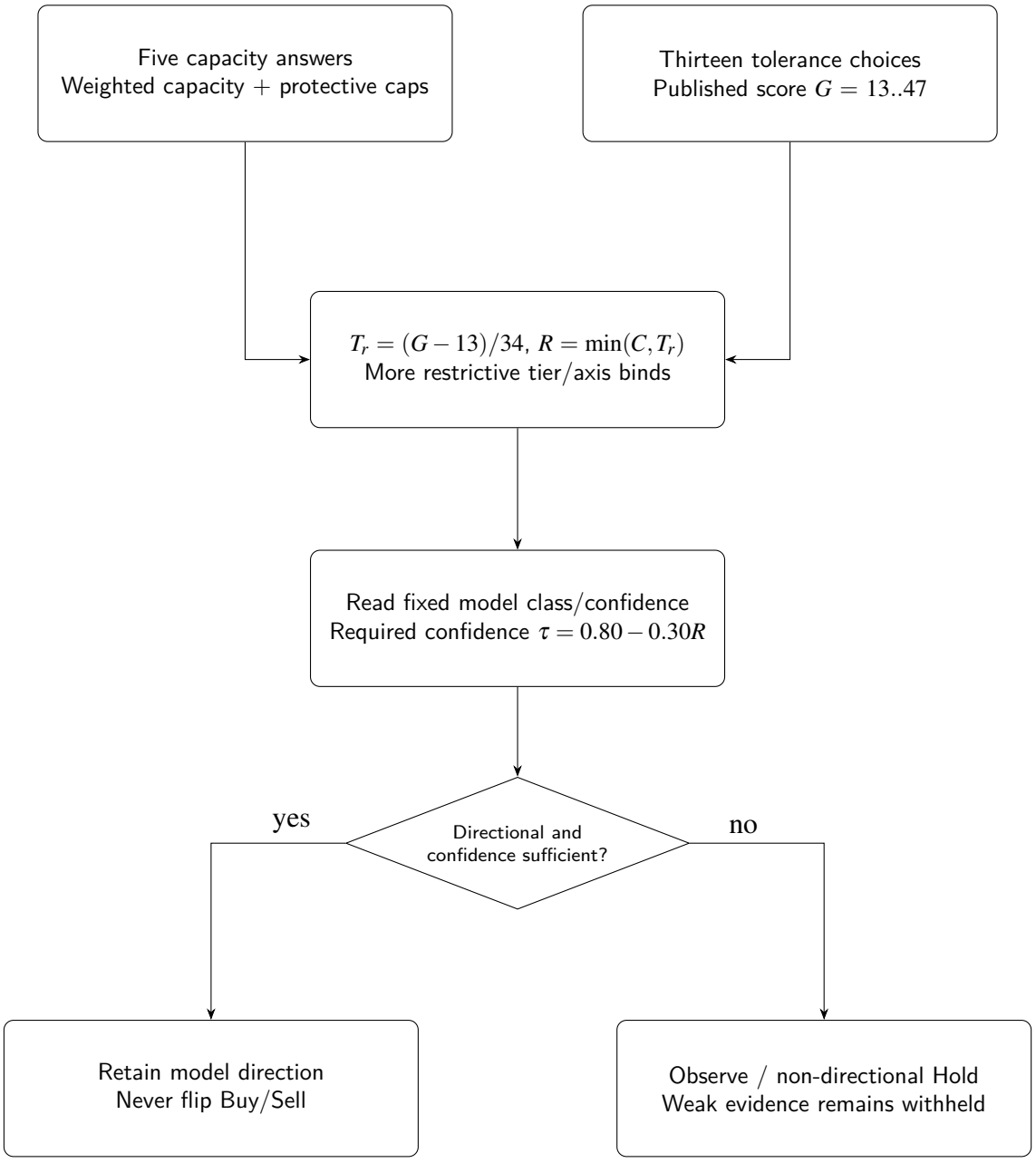


Figure 6.2: Risk capacity/tolerance calculation and direction-preserving confidence policy.

6.9 Demonstration Evidence

The route is Overview, Onboarding, Decision, Evidence, Risk profile and Explanation check. Onboarding explains purpose, loss risk and the absence of trades. An abstaining Decision headlines “No recommendation (Observe)” and shows Buy/Hold/Sell only as diagnostic evidence. Setup is moved to Appendix D.

Table 6.6: Supported deliberate corruption and expected recovery.

Corruption	What is changed	Expected result
Wrong value	Recorded feature value in text/payload	Reject contradictory value
Wrong sign	SHAP support/opposition or contribution sign	Reject wrong selected-class direction
Wrong rank	Position in top-k attribution order	Reject inconsistent rank
Unknown feature	Feature/entity absent from record	Reject unsupported feature
Displayed probability	Text says 1.00% beside correct JSON	Reject prose independently; checked fallback restores record
Wrong policy	Observe changed to Buy in prose or JSON	Reject against trusted policy; fallback stays Observe

These supported fixtures are generated by `explain/fixtures.py` and exercised in the portable/policy tests. The visible 1.00% button is one example of that wider test scope. Edits and case/profile changes remove stale verification.

6.9.1 Current Interface and Actual Captures

Only the selected step runs its widgets. Decision shows the policy result and probabilities. Evidence shows a feature value, maintained definition and selected-class SHAP effect. Risk profile shows capacity, the tolerance total, binding constraint and required confidence. Explanation check shows the candidate, claim report and separately verified output. Each step includes historical date/horizon, confidence and a short interpretation note.

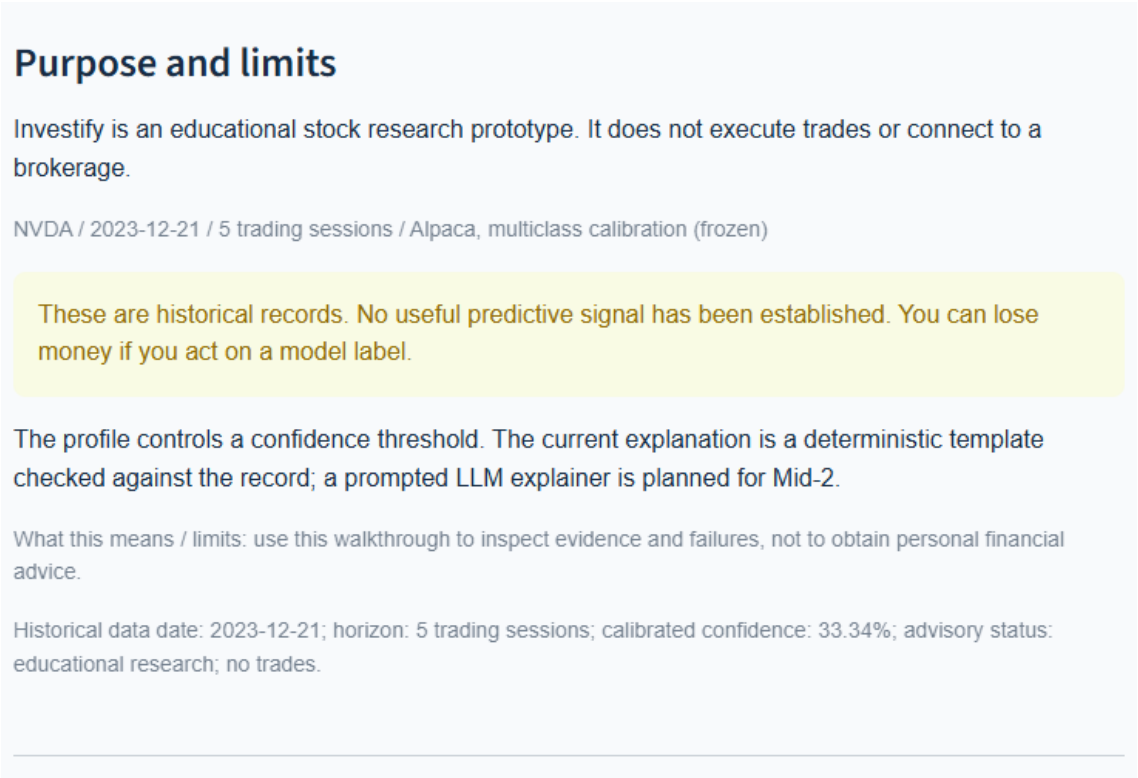


Figure 6.3: Current onboarding purpose, risks, date/horizon and confidence.

The captures below are actual views from the revised browser journey. They show the current Observe headline, profile threshold and policy-aware fallback; the capture receipt records their source files and image hashes.



Figure 6.4: Current decision panel with the full Observe headline and a secondary diagnostic class.

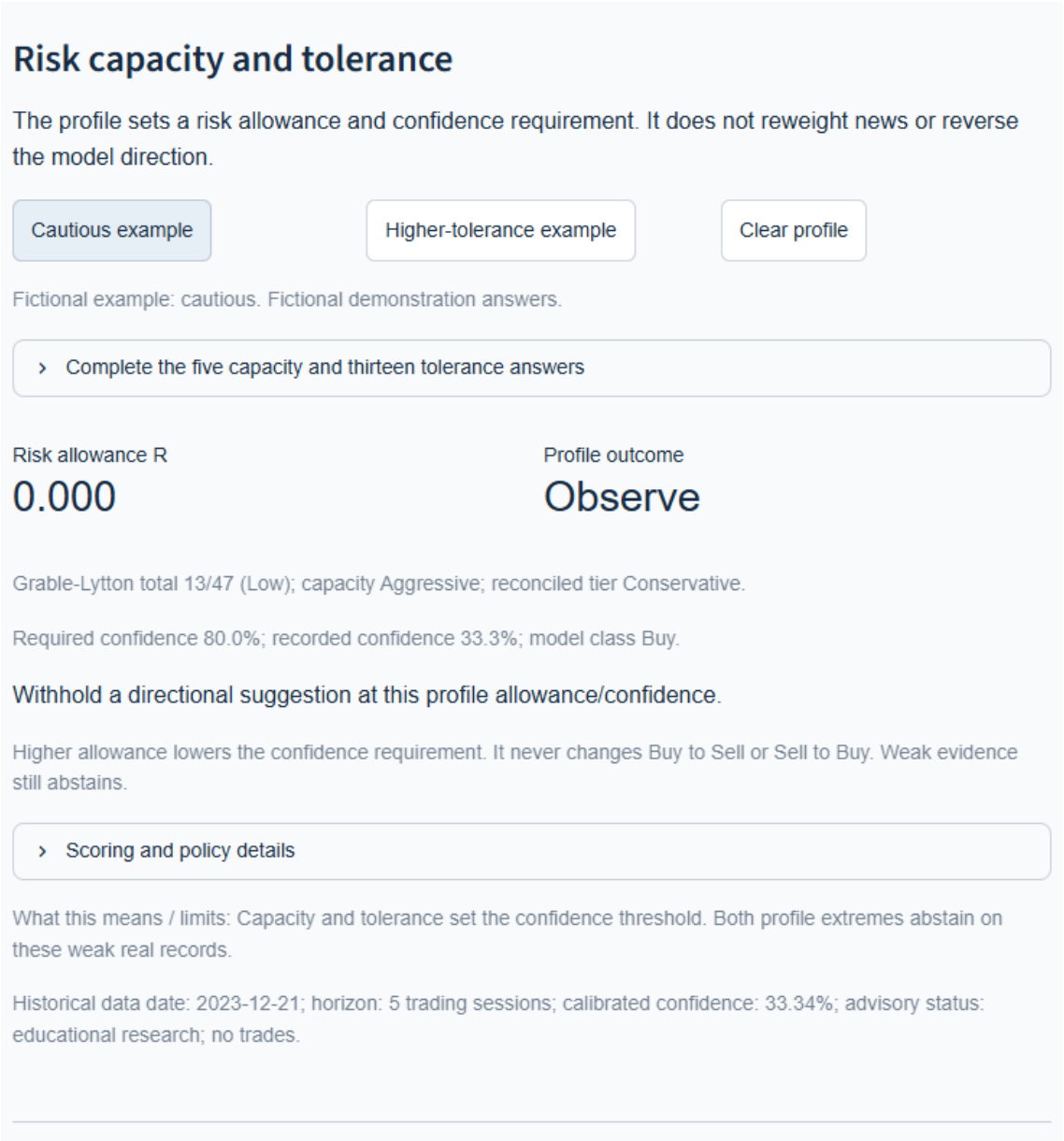


Figure 6.5: Current cautious profile: tolerance 13/47, allowance zero and 80% required confidence.

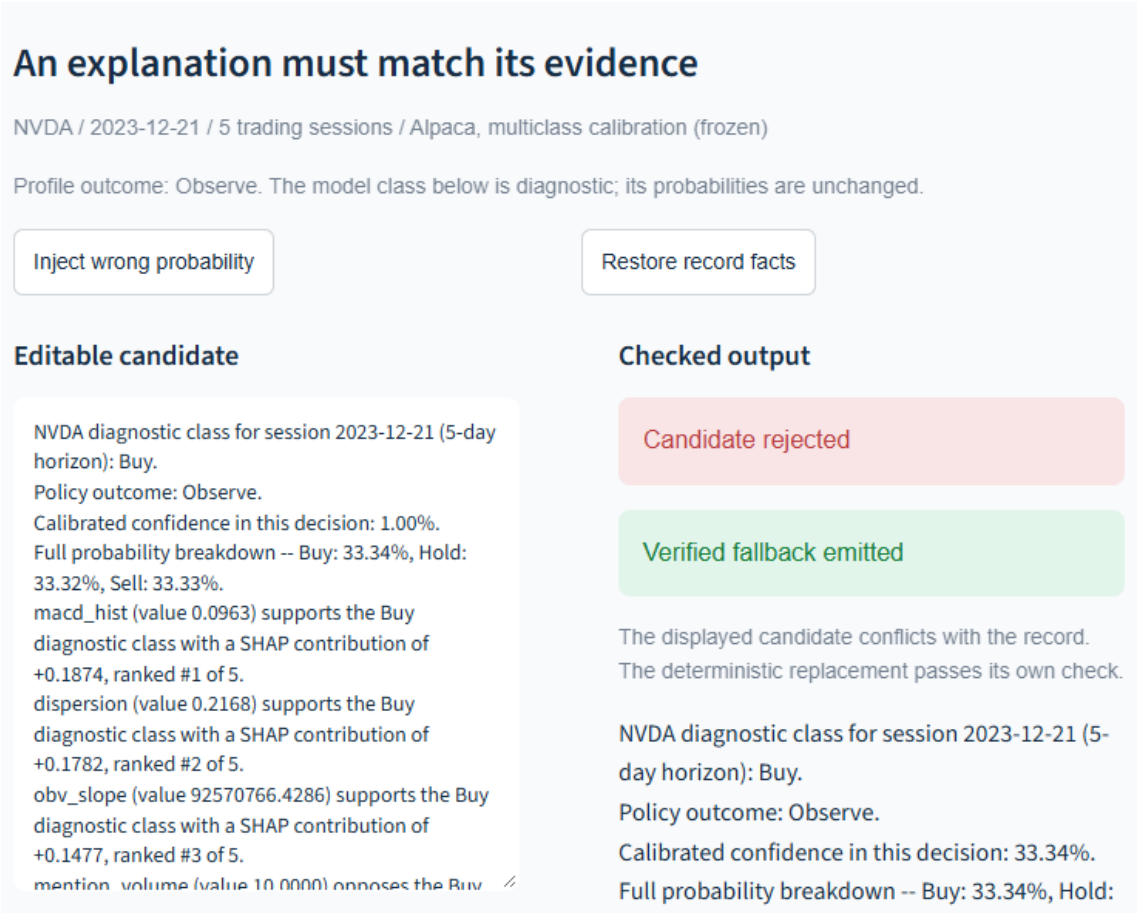


Figure 6.6: Current displayed-probability corruption and checked Observe fallback.

6.10 Iteration Outcome

The live explanation/profile module is implemented and locally tested. The scientific result is weak prediction and a failed useful-confidence criterion. The remaining LLM, divergence, service, NFR and participant work follows the staged plan in Chapter 9.

Chapter 7

Testing and Early Validation

7.1 Testing Strategy

Pair expected behavior with deliberate errors and retain exact logs. Software tests check consistency and failure handling; scientific evaluation checks prediction/calibration on declared populations. Neither is a participant study.

7.2 Test Levels and Techniques

Unit tests cover record invariants, formulas, omissions and malformed input. Module tests combine context/profile/template/checker. Streamlit AppTest exercises widgets and state; localhost HTTP checks exercise the portable service. Receipt and extraction guards check source/PDF packaging. Scientific fault arms and authored classifier probes have their own population and interpretation.

7.3 Test Cases

Appendix C lists input, expected behavior and named checks. The live demonstration emphasizes probability corruption, while fixture tests also cover values, signs, ranks, unknown features, missing claims, wrong policy and stale state.

7.3.1 Evaluation Metrics and Targets

Table 7.1: Definitions, targets and current measurement scope.

Metric	Definition / target	Scope and status
Numeric fidelity	Correct verifiable numeric claims / verifiable numeric claims; $\geq$ 0.98	Template/corruption checks executed; generated-model held-out evaluation planned
Directional consistency	Correct SHAP direction claims / verifiable direction claims; $\geq$ 0.95	Selected-class effects; not price causation
Claim recall	Correct displayed expected slots / all expected slots; ≥ 0.80	Essential displayed claims also have a hard missing-claim gate
Unverifiable-claim rate	Unsupported claims / all displayed claims; ≤ 0.15	Current strict UI rejects every unparsed line
Contradictory emission	Known contradictory emitted claims; target zero across 200 held-out explanations	The proposed 200 generated explanations are not completed
Macro-F1 and lift	Mean of class F1, $F1_k = 2P_kR_k/(P_k + R_k)$; lift versus random and linear	Six-fold NVDA and final 186 cases measured; Gate B not passed
Brier and reliability	$N^{-1}\sum_i\sum_k(p_{ik} - y_{ik})^2$; bin predicted confidence versus observed frequency	Brier measured; lower is better; comprehensive reliability assessment remains
Social macro-F1/recall	Mean class F1; $R_k = TP_k/(TP_k + FN_k)$	Original classifiers on frozen 24-post held-out references
Annotation kappa	$\kappa = (p_o - p_e)/(1 - p_e)$ for independent raters	No independent second rater; no measured kappa
User study	Completion rate, time, assistance, interpretation errors and top-driver identification	Participant evaluation planned; no invented success target/result
Runtime NFRs	p95 latency, peak memory and documents/s	Proposal environments and targets in Table 4.6; unmeasured

Raw generated output and filtered emitted output must be scored separately; a fallback must not hide a weak generator. Numeric rounding tolerances are defined in the verifier and are separate from whole-set targets.

7.4 Test Results

The local Mid-1 runner executes both jobs from `code/` into new evidence directories. Its actual unittest logs report the following scopes. They overlap with individually selected tests and must not be summed with earlier receipts.

Table 7.2: Observed local unittest scopes for this revision.

Scope	Actual "Ran N tests"	Result
CI source/metadata guards	7	Pass
Portable `test_mideval*.py`	183	Pass
Legacy Streamlit	24	Pass
Current presenter	19	Pass
Recorded-study selectors	4	Pass

Ruff, record validation, preview/plan-only, risk comparison and real localhost HTTP checks also passed; they are not additional unittest counts. New final receipts retain the exact commands, source digests and logs. CI restores the original tracked source-metadata bytes after execution.

7.4.1 Social Results and Reference Construction

The saved study acquired 200 MSFT posts, retaining 117 references and 83 exclusions. The chronological split is 93 development/24 held out; held-out classes are 7 positive, 15 neutral and 2 negative. CryptoBERT macro-F1 is 0.2564 with exact agreement 15/24, all predictions neutral. Cardiff macro-F1 is 0.6016 with exact agreement 16/24, positive recall 4/7 and negative recall 1/2.

References were constructed in one model-assisted pass declared as OpenAI/Codex GPT-6.1 Sol configuration, without an independently authenticated serving revision. No human annotation or inter-rater agreement was measured. Scores describe agreement with those references. Five separate authored CryptoBERT probes produce positive and negative outputs; the nominal neutral probe is negative. These diagnostic probes do not replace the held-out benchmark.

7.5 Acceptance Testing / Validation

The engineering acceptance path is case selection, evidence, complete profile, policy, checking and recovery. The transparency tests verify onboarding, captions and an Observe headline. Participant acceptance, independent external installation and formal supervisor review have not been observed.

7.6 Edge and Failure Cases

Missing/malformed records withhold replacement decisions. Incomplete profiles withhold scoring. Nonfinite values, unknown features, missing essential claims, conflicting prose/JSON and wrong policy fail. Changing the source/session/profile/text clears the verdict. Failed fallback is surfaced as an error. Real uniform-confidence cases remain Observe under both example profiles; controlled 0.70 fixtures exercise qualifying directional cases separately.

7.7 Defects and Corrective Actions

Corrections include payload-over-prose acceptance, omitted evidence, stale state, label-end purging, unavailable calibration labels and portable provenance paths. This revision aligns the Observe headline and explanation policy, adds onboarding/step disclosures and fixes an indentation error left in a UI test. Template and verifier patterns change together, while original teacher packets remain replayable.

7.8 Current Validation Status

The local checks establish the bounded software path. The final hosted matrix and delivery are identified in Appendix D once published. Predictor usefulness, generated prose, population-specific profiling and human acceptance are evaluated separately in the remaining plan.

Chapter 8

Project Management and Development Evidence

8.1 Work Completed

Delivered work includes the historical data/harness, recorded engine/attribution outputs, complete evidence context, scored profile, deterministic verification/recovery and CI source/PDF delivery. The report and UI revision makes policy outcome and template/LLM status consistent.

8.2 Work Remaining

Integrated divergence, multi-company coverage, useful prediction/calibration, the prompted explainer, constrained free-prose claims, measured NFRs, parity/degraded service, QLoRA and user acceptance remain. These items are scheduled rather than counted as implemented.

8.3 Individual Responsibilities

Ahmed Javaid owns data/timing and harness/CI; Rafay Ul Haq owns UI/profile and sentiment integration; Soban Ahmad owns prediction/calibration/SHAP and verification/student work. These follow the proposal allocation and the user's confirmed delivery account.

8.4 Contribution of Each Team Member

Table 8.1: Delivered work and continuation responsibility.

Member	Delivered part	Continuing work
Ahmed Javaid	Calendar/joins, label/purge, leakage evidence and CI/delivery	Multi-company timing, parity and deployment
Rafay Ul Haq	Interface, risk questionnaire, sentiment study and demo preparation	Divergence, degraded operation and acceptance
Soban Ahmad	Engine/calibration/SHAP, verifier/fallback and teacher data preparation	Baseline/calibration review, prompted student and QLoRA

The user confirmed equal one-third delivered contributions. The revised fourteen-slide talk allocates 220 seconds per member, including the shared live demonstration, for 660 seconds total. Speaking assignments and code preparation are in the team guide.

8.5 Repository and Version Control Evidence

Repository: https://github.com/Ahmed-Javaid/Investify-FYP-2027. The existing feature branch is preserved. Appendix D identifies the published code snapshot, successful Actions run and revision tag. A document cannot contain its own eventual commit hash without a later document commit; the cited implementation snapshot and subsequent delivery commits are distinguished explicitly.

8.6 Milestones and Commit Evidence

The retained scientific run records its execution date and source identities. Local history includes checkpoint `5f6b6ff` before the authorized cleanup, `db4a3e8` for consolidated documentation cleanup, and `42fd42c` for retained-figure source-delivery verification. These are historical milestones, not proof of the new uncommitted revision. Relative future milestones are in Table 3.4.

8.7 Tools and Tool-Assisted Development

Codex and Claude Code were used for coding assistance, test construction and document revision. The recorded explanation teacher is the OpenAI GPT-6 assistant; exact serving revision, token usage and example latency are not recorded. Social reference construction separately declares GPT-6.1 Sol session configuration. The team remains responsible for reviewing and explaining the submitted work.

8.8 External Components and Dependencies

FinBERT, CryptoBERT/Cardiff, XGBoost, TreeSHAP, the Grable–Lytton scoring instrument, market calendars, Streamlit, Python/LaTeX and GitHub Actions are external inputs. Project work concerns their timing, integration, evaluation, error handling and reporting. Model/domain/license status is in Table 2.3; original data/models are identified in the saved receipts.

8.9 Risks and Mitigation

Table 8.2: Prospective project risk register adapted from concept Section 17.

Risk	Likelihood	Impact	Mitigation	Owner
Silent temporal leakage	Medium	High	Prior joins, label endpoints, fold-local fits and permanent canary	Ahmed
Engine fails baselines	High	High	Report null result; preserve controls; development-only analysis	Soban
Control shows apparent signal	Medium	High	Halt, audit and distinguish fault geometry from skill	Ahmed
Historical data differs from deployment year	High	Medium	Bound the historical window and separately identify future data	Ahmed
Social model fails equity transfer	Medium	Medium	Paired frozen benchmark; news-only fallback; independent references later	Rafay

Explanation comparison saturates after gating	Medium	Medium	Score raw output before guard and emitted output separately	Soban
Deployment delayed	Medium	High	Early container/parity work before final milestone	Ahmed
Cross-owner integration failure	Medium	High	Shared interfaces, end-to-end checks and same-source CI	All
Local GPU below proposed training budget	High	Medium	Measure actual device; select a verified compatible run or revised budget	Soban

Likelihood and impact are planning judgements, not measured frequencies.

8.10 Security and Privacy Considerations

The demo binds to loopback, stores profile answers in session memory and connects to no brokerage. Private `code/.env` credentials and external weight/cache files are excluded from delivery. Production authentication, abuse cases, retention rules and consent require future service/participant work. Software dependency licenses do not automatically authorize source-data redistribution.

8.11 Current Limitations

1. The prediction study covers NVDA and one historical interval. Broader companies, regimes and continuous-market generalization are unmeasured.

2. Source timing rules are implemented, but historical vendor availability and a byte-identical original acquisition are unproven. The fixed 16:00 cutoff still needs early-close reconciliation; cash dividends are not adjusted.

3. Clean prediction is below stratified random. Isotonic collapses classes; temperature reaches 1000 and approximately uniform confidence. No useful forecasting or return claim follows.

4. Risk is a post-hoc confidence gate, not a fitted personalization feature. All real temperature cases remain Observe. Local questionnaire wording and suitability have not been population-validated.

5. Social signals are not integrated into NVDA features. The frozen 24-post MSFT comparison uses a single AI reference pass; human validity and kappa are unavailable.

6. The implemented explainer is deterministic. The verifier recognizes template grammar, not unrestricted semantic/causal claims. Six teacher examples do not establish a trained or useful student.

7. Runtime NFRs, production hosting, live/backtest parity, independent installation and participant comprehension are not yet measured in the proposed environments.

8. Submission date/session, signatures and formal meeting records have not been supplied. The corresponding fields remain blank; no institutional approval is inferred.

Chapter 9

Remaining FYP Plan

9.1 Remaining Scope

Complete the broader stock research workflow, evaluate its contributions and document its operational boundaries. Brokerage execution remains excluded.

9.2 Remaining Modules and Features

Add audited company coverage, paired news/social features and divergence, a service layer, prompted explanations with constrained claims and optional verified student adaptation. Review the profile wording and the actual-close timing policy.

9.3 Remaining Iterations

Iteration 2 completes engine/service and the preregistered divergence comparison. Iteration 3 (Mid-2) adds the prompted explainer, free-prose claim binding, NFR/parity/degraded-operation measurement and acceptance preparation. Iteration 4 compares original and QLoRA-adapted Phi-3-mini with the template and performs broader evaluation and handover.

9.4 Implementation Plan

First audit additional source coverage and available-label boundaries. Freeze the divergence protocol before evaluating variants. Load a verified original student checkpoint,

score its raw prompted baseline, expand filtered examples and preserve the held-out set before adaptation. Integrate the tested service and interface incrementally.

9.5 Testing and Validation Plan

Repeat the timing/canary controls for each scientific identity. Compare macro-F1, baseline lift, Brier/reliability and class distributions under shared splits. Measure raw and gated explanation quality. Inject missing feeds/thin history and pin a historical clock for serving parity. The automated workflow is **2 jobs $\times$ 2 OS = 4 checks, then 1 gated delivery job**; local counts remain distinct from classifier or participant evidence.

9.6 Deployment Plan

Prepare a container and an explicit target VM, measure the required environments and deploy the research service. Test installation from the documented package on a separate machine. Current CI delivers downloadable source/PDF artifacts; it does not deploy a production service.

9.7 Tentative Timeline

Use the proposal's relative checkpoints in Table 3.4. Formal assessment appointments have not been supplied. Engine/divergence/service work precedes the Mid-2 prompted explainer, and QLoRA follows a working prompted baseline.

9.8 Expected Final Deliverables

Deliver integrated source/service, source-bound experiments and ablations, timing/calibration and explanation evaluations, measured performance/degraded-operation evidence, external installation instructions, participant records where completed, and final report/presentation. Retain a defensible null-result report if the learned engine has no useful advantage.

Chapter 10

Conclusion

Investify Mid-1 implements an inspectable historical research flow with timing controls, complete evidence, a scored profile, deterministic explanation checking and verified recovery. The retained experiments show weak prediction, an isotonic class collapse and temperature confidence near one-third. The interface therefore presents Observe rather than a directional recommendation.

The revision responds to PF-01–PF-09 through explicit contributions, a scheduled LLM design, model/metric tables, market background, transparency acceptance and a demonstrable core. The next work is integrated divergence, useful prediction/calibration evaluation, the Mid-2 prompted explainer and operational/user validation. QLoRA remains Iteration 4.

Appendix A

Proposal Action Register

The full feedback/action/status/evidence table is in Section 3.2. The preserved original panel comments and interpretation are in `sources/Mid_Eval/PROPOSAL_ACTION_REGISTER.md`; the supplied decision is "Approved with Minor Modifications."

The approved presentation is `sources/InvestifyUpdated(1).pptx`. Its slide 8 separates the harness/verifier/divergence claims from imported algorithms and excludes profit claims. The concept document is reference material. Table 3.3 records its Iteration-1 items and deviations rather than presenting them as completed approved scope.

Appendix B

Detailed Requirements Traceability

The versioned matrix maps requirements to actual source symbols and named regressions. It verifies implementation/test anchors; scientific quality and participant acceptance remain separate.

Table B.1: Functional requirement, implementation and regression traceability.

ID	Implementation	Named check
MID-FR-01	`DecisionRecord.from_dict`	`test_nonfinite_out_of_range_or_boolean_probabilities_are_rejected`
MID-FR-02	`build_context`	`test_all_real_records_keep_their_facts_and_validate`
MID-FR-03	`verify`	`test_correct_payload_cannot_hide_wrong_displayed_probability`
MID-FR-04	`verify`	`test_payload_cannot_supply_facts_or_notice_absent_from_displayed_text`
MID-FR-05	`emit`	`test_fallback_retains_the_rejected_candidate_report`
MID-FR-06	`verify`	`test_strict_mode_rejects_unparsed_text_and_preserves_clean_template`
MID-FR-07	`harness_input_report`	`test_empty_partition_directories_do_not_enable_a_rerun;test_leakage_view_labels_saved_results_and_limits_preflight_claims`

ID	Implementation	Named check
MID-FR-08	`LocalExplainer.explain;verify_runtime`	`test_unavailable_provider_fails_fast_with_explicit_fallback;test_all_shards_are_required_and_changed_bytes_fail_before_inference`
MID-FR-09	`verify_generated`	`test_wrong_reviewed_definitions_in_text_or_json_are_rejected`
MID-FR-10	`evaluate`	`test_plan_drift_is_rejected_before_any_provider_request`
MID-FR-11	`verify_generated`	`test_every_feature_has_a_value_magnitude_direction_and_target_in_both_sources`
MID-FR-12	`score_profile`	`test_incomplete_or_boolean_tolerance_answers_are_refused;test_capacity_caps_bind_despite_high_tolerance`
MID-FR-13	`build_comparator`	`test_changing_heldout_feature_values_cannot_change_scaling_statistics`
MID-FR-14	`apply_risk_policy`	`test_risk_allowance_changes_abstention_without_reversing_direction`
MID-FR-15	`validate_benchmark`	`test_one_ai_pass_has_null_agreement_and_explicit_reference_scope`
MID-FR-16	`evaluate_predictions`	`test_no_runs_or_one_candidate_never_manufactures_a_model_selection`
MID-FR-17	`run_inference`	`test_actual_loader_refuses_missing_weights_without_a_download`
MID-FR-18	`import_responses`	`test_actual_exports_keep_validation_out_of_training_and_disclose_hybrid_targets;test_actual_teacher_view_replays_examples_and_preserves_student_pending_state`

ID	Implementation	Named check
MID-FR-19	`fit_temperature;softmax_temperature`	`test_calibration_refuses_unavailable_labels;test_real_calibration_replay_preserves_all_raw_classes_and_probabilities`
MID-FR-20	`score_profile`	`test_thirteen_published_item_scores_have_exact_minimum_and_maximum`

The machine-readable matrix is `code/docs/requirements_traceability_v2.json`. The earlier eighteen-ID matrix remains preserved.

B.1 Transparency Acceptance Anchors

MID-FR-21 uses `test_onboarding_precedes_the_decision_and_discloses_purpose_risks_and_status`. MID-FR-22 uses `test_each_walkthrough_step_has_advisory_date_horizon_and_limits`. MID-FR-23 uses `test_walkthrough_renders_only_the_selected_step`, `test_abstaining_explanation_and_fallback_keep_policy_outcome`, and independent prose/payload policy tests.

B.2 Proposal FR-1–FR-15 to Mid-1 Mapping

Table B.2: Reference FR acceptance and delivered boundary.

ID	Reference requirement	Mid-1 mapping	Status / remaining work
FR-1	Append-only dual timestamps/dedup	MID-FR-01, 07; store/ingest tests	Partial: code exists; historical availability unproven
FR-2	Session and continuous-window mapping	MID-FR-07; calendar tests	Partial: code/tests; broader calendar study pending
FR-3	Strict backward as-of join and controls	MID-FR-07; causality/split controls	Partial: Gate A not fully closed
FR-4	Training-fold normalization	MID-FR-13; scientific scaling	Implemented; comparator and engine scopes separate

FR-5	Calibrated five-session decision	MID-FR-01, 19	Partial: class/probability exists; useful calibration failed
FR-6	Additive TreeSHAP	MID-FR-02; native replay/additivity	Implemented on retained records
FR-7	Restrictive capacity/tolerance tier	MID-FR-12, 20	Implemented scoring; population validation pending
FR-8	Verified explanations over 200 cases	MID-FR-03–05, 23	Partial: template guard; generated 200-case study pending
FR-9	Per-claim score/report	MID-FR-03–05	Implemented for supported claims
FR-10	Controls on every evaluation and CI	MID-FR-07; canary suite	Partial: portable CI does not rerun full science
FR-11	Pinned-clock live/backtest parity	No completed Mid-1 acceptance	Open: service parity study
FR-12	Feed/history degraded state	MID-FR-01, 07, 11	Partial: saved-input refusal; live-feed fault injection pending
FR-13	Persist source/model/data identity	MID-FR-01, 11, 18, 19	Implemented saved records/replay; future service persistence
FR-14	Notice/horizon on every response	MID-FR-21–23	Implemented current walkthrough; future service paths pending
FR-15	Install by a person outside team	Setup and extraction checks	Partial: automated checks; independent person not observed

Appendix C

Detailed Test Cases

Inputs and observations below distinguish real retained cases from explicit diagnostic fixtures. Final execution receipts identify the current source snapshot and test scopes.

Table C.1: Individual input, expected-result and actual-result test cases.

ID / case	Purpose	Input	Expected	Actual
TC-01	Valid NVDA case	2023-12-21, multi-class frozen	Valid source/date record; no replacement	Loaded; probabilities normalized
TC-02	Malformed input	Missing/invalid record	Error and no invented class	Error/refusal regression passed
TC-03	Displayed probability corruption	1.00% prose beside correct JSON	Reject candidate; checked fallback	Rejected; fallback accepted
TC-04	Structured contradiction	Correct prose, incorrect JSON claim	Independent payload rejection	Rejected in verifier regression
TC-05	Empty/omitted display claim	Missing decision or notice	No acceptance from payload alone	Rejected by hard coverage gate
TC-06	Unsupported strict sentence	Unparsed market prose	Strict rejection	Rejected; clean template accepted
TC-07	Wrong feature value/direction	Modified supported claim	Reject mismatch against record	Value and sign checks passed
TC-08	Fallback failure	Template fails verification	Error, no success claim	Failure path regression passed

ID / case	Purpose	Input	Expected	Actual
TC-09	Edited candidate	Edit after accepted verdict	Old verdict disappears	Widget invalidation passed
TC-10	Changed case	Select another session	Reset candidate/result identity	Widget case-change passed
TC-11	Incomplete profile	Only three tolerance answers	Refuse incomplete eighteen-answer form	ValueError; no profile
TC-12	Minimum tolerance	All published minimum choices	Total 13, allowance 0	13; R=0.000
TC-13	Maximum tolerance	Maximum choices, full capacity	Total 47, allowance 1	47; R=1.000
TC-14	Protective capacity cap	No buffer, high tolerance	Capacity binds below 0.4	Conservative; cap binds
TC-15	Direction preservation	Controlled Buy/Sell confidence 0.70	Low R abstains; high R retains direction	Low: Observe; high: original class
TC-16	Real uncertainty	Near-uniform corrected records	No unsupported directional suggestion	Hold remains neutral; others Observe
TC-17	Calibration availability	Outcome ends at evaluation start	Refuse fit	ValueError before fit
TC-18	Multiclass replay	186 margin vectors per mode	Same raw argmax; sum 1	All rows preserve order
TC-19	Extreme margins	10000, 9999, -10000	Stable finite normalized output	Passed; first class remains largest
TC-20	Invalid temperature	Zero, negative, boolean, NaN	Refuse calibration parameter	All invalid values refused
TC-21	Study partial output	Failed/missing candidate rows	No paired score substitution	Coverage refusal passed
TC-22	Original CryptoBERT probes	Five authored crypto sentences	Outputs vary; retain raw results	Both directions; neutral probe failed
TC-23	Delivery admission	Failed, stale or wrong-commit receipt	Do not deliver	Nine guard cases passed
TC-24	Checkout content drift	Changed bytes beyond newline translation	Refuse restoration	Content and path refusal passed

TC-15 and TC-19–20 are controlled fixtures; TC-22 is a model diagnostic. Suite receipts remain separate.

Appendix D

Repository and Contribution Evidence

D.1 Repository, Revision and CI Evidence

Repository: https://github.com/Ahmed-Javaid/Investify-FYP-2027. The authorized publication uses the retained feature branch, a reviewed pull request and tag `mid1-v1.1`. The exact pushed implementation commit and successful same-commit Actions run are inserted after the hosted check; this pre-publication build claims only the completed local checks. Published implementation commit and successful Actions run: not yet available for this working revision. The source is on `feature/mid1-report-revision`; local evidence and UI jobs pass. This block is replaced by observed commit/run details after publication.

D.2 Demo Setup

Use Git and Python 3.11. Create a separate recorded-viewer environment and install `code/requirements-demo-runtime.txt`; do not install the scientific `pyproject.toml` into it. From the repository root, restore only documented newline translations with `tools/restore_ci_checkout.py`, then enter this command on one line:

```
python -m streamlit run code/app/streamlit_app.py
 --server.address=127.0.0.1 --server.port=8507
 --theme.base=light --server.headless=true
```

Open http://127.0.0.1:8507 and keep the server terminal open. The prepared laptop can use `tools/start_guided_demo.ps1`. On another machine use the documented Python path. If the port is occupied, choose another port; do not stop an unrelated process. The team guide supplies the click script and troubleshooting.

D.3 Scientific Reproduction

The original Alpaca study remains in `code/artifacts/scientific_runs/alpaca_revision_conservative_20261012_run01`. The temperature diagnostic is separately identified in `code/artifacts/scientific_runs/temperature_calibration_20261013_run01`. Replaying saved records is distinct from reacquiring source data or rerunning original publisher inference. Social plan, reference labels, checkpoints and original outputs remain unchanged.

D.4 Local CI Commands

From `code/`, choose new output directories:

```
python -B scripts/run_mid1_ci.py --job evidence
 --output ../evidence/NEW_evidence
python -B scripts/run_mid1_ci.py --job streamlit-ui
 --output ../evidence/NEW_ui
```

Each command is one line. Run shared local jobs sequentially because they temporarily declare source identity and restore the prior tracked bytes. Hosted Windows/Ubuntu jobs run in independent checkouts. Four passing same-commit checks gate the fifth delivery job.

D.5 Document Control and Disclosures

Table D.1: Document control for the revised submission.

Field	Value
Project	Investify, F26-003
Assessment	FYP-1 Mid-Evaluation, Fall 2026
Submission date	_____
Batch/session	_____
Version	1.1
Authors	Ahmed Javaid, Rafay Ul Haq, Soban Ahmad
Supervisor	Mr. Irfan Ullah; no co-supervisor
Source authority	Official F26 template/rubric and approved proposal slides; concept is reference

The supplied official template PDF was available; the named template ZIP is not present in the current workspace. The unchanged institutional class/logo and headings were checked against that PDF. Coding assistants and the teacher/reference construction are disclosed in Chapters 7–8. Blank date/session/signature fields and the empty meeting form are intentionally unfilled.

D.6 Build and Delivery

Compile report and Beamer sources in fresh E: scratch directories, process citations, render every page and inspect the output before selecting new deliverables. New source/PDF receipts identify the reviewed outputs. CI source delivery checks those identities and performs isolated extraction. Earlier receipt JSON and scientific outputs remain unchanged.

Appendix E

Additional Design Diagrams

The main use-case, architecture, domain/schema, sequence and deployment diagrams are in Chapters 4–6. This appendix contains supporting data-flow, activity, state, layering and future-wireframe models. All are editable TikZ sources.

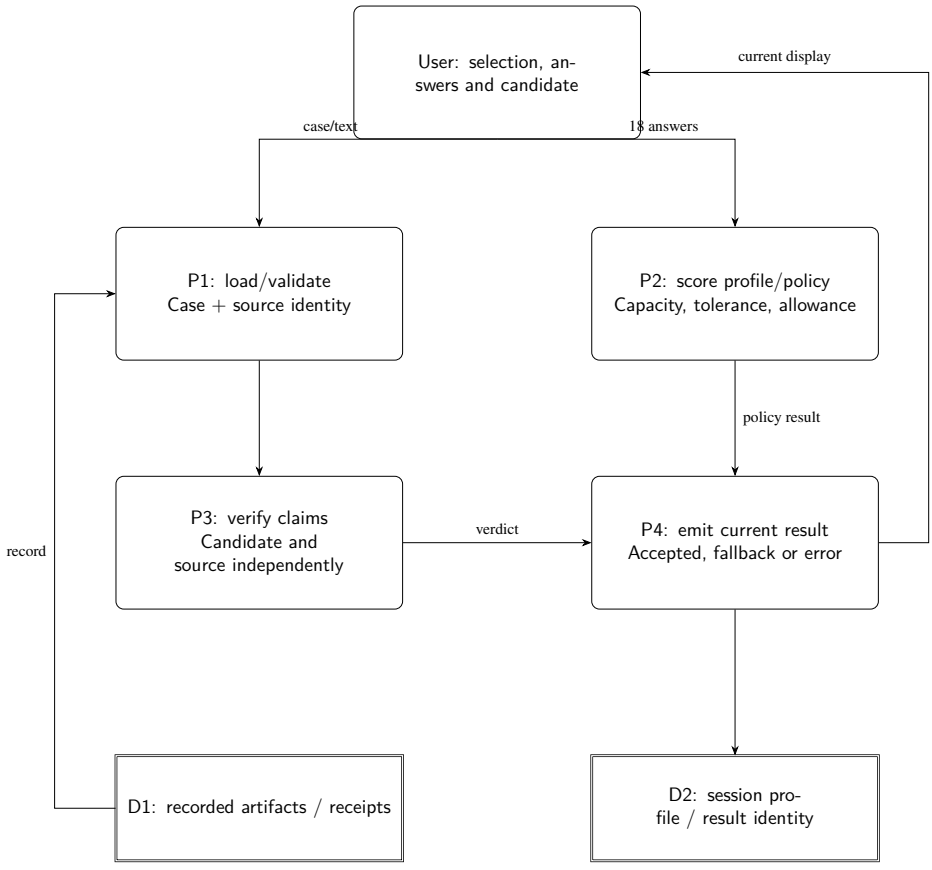


Figure E.1: Data flow through case, profile and explanation checks.

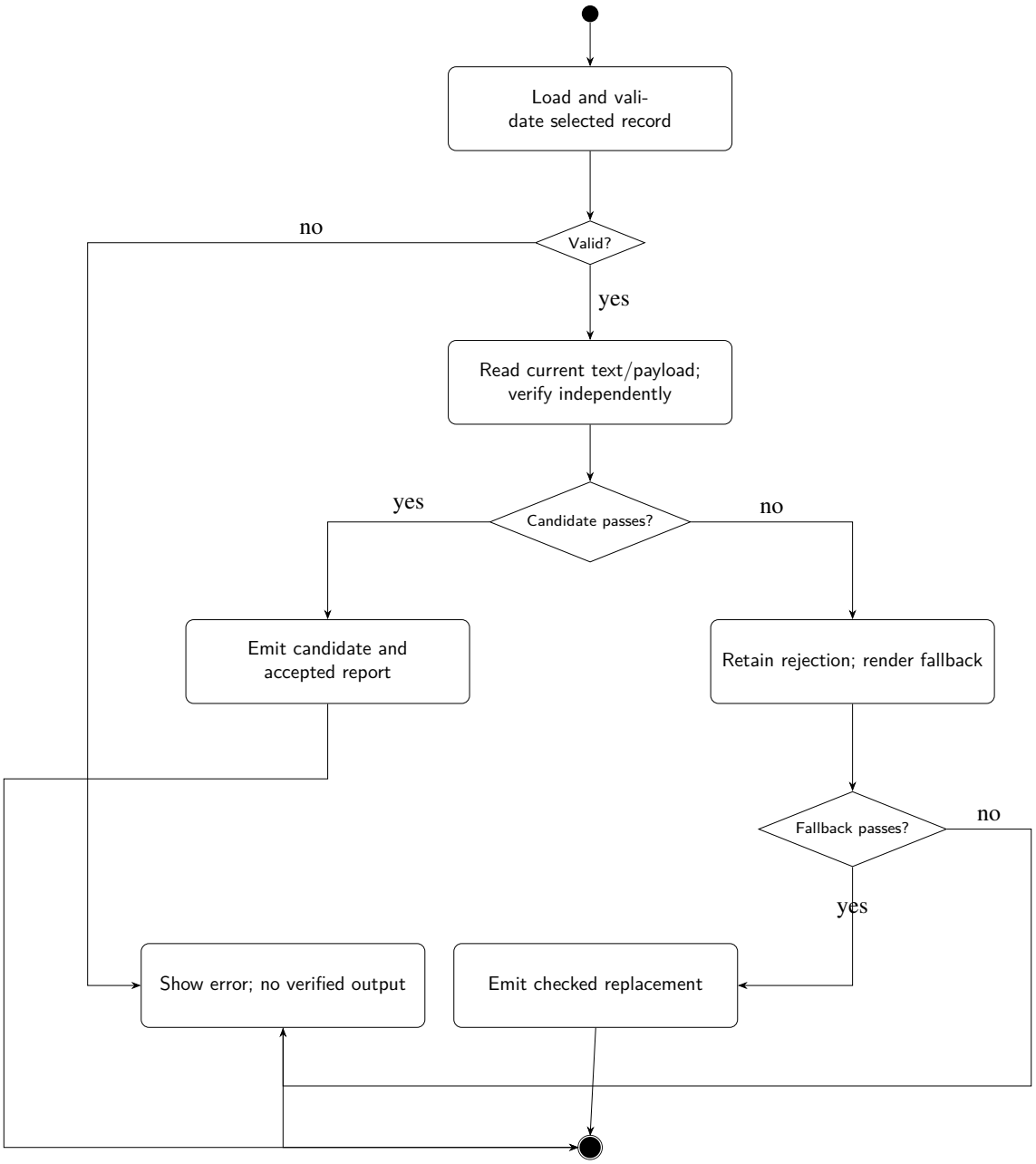


Figure E.2: Checking activity and failure branches.

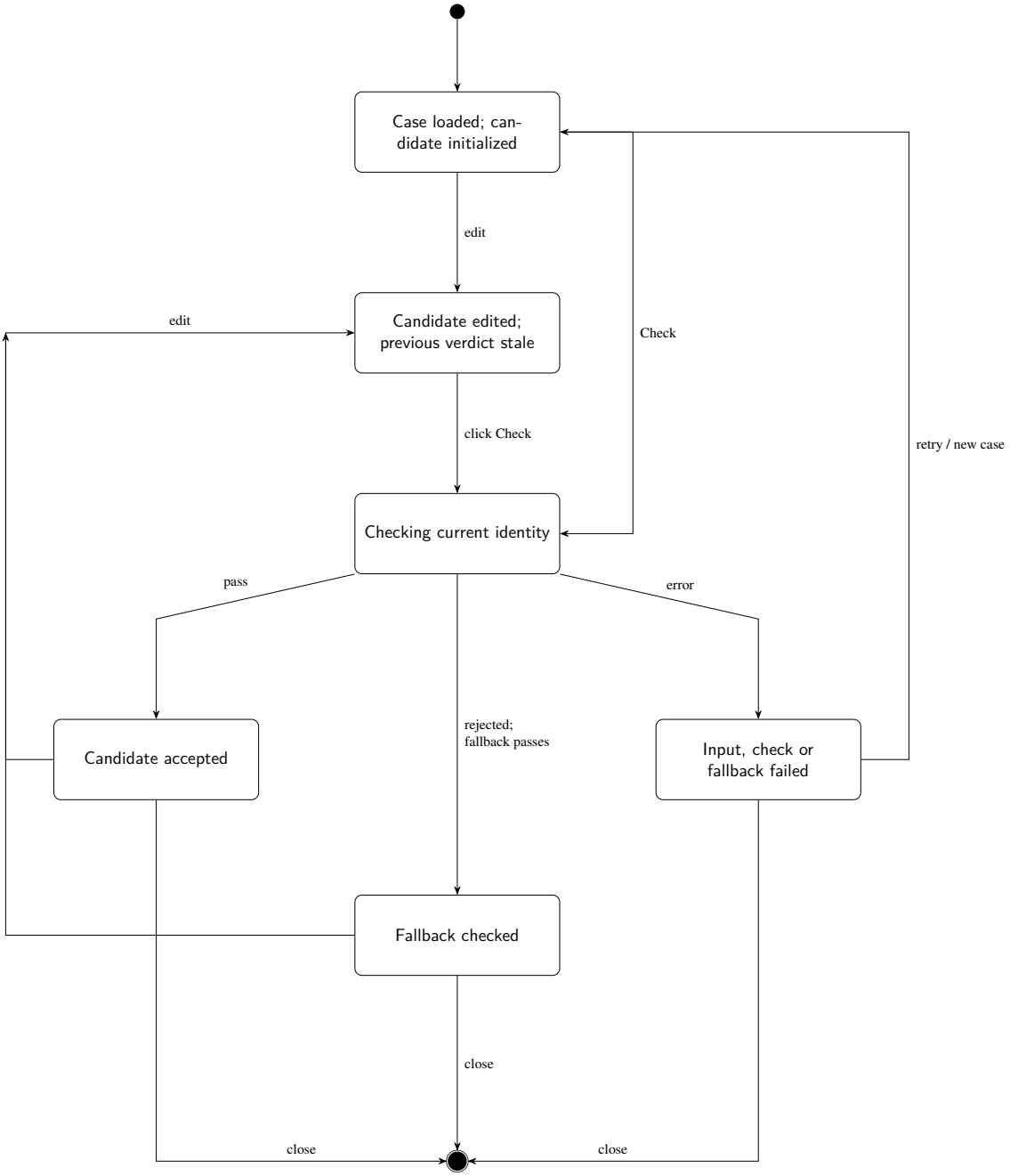


Figure E.3: Verification state and stale-result invalidation.

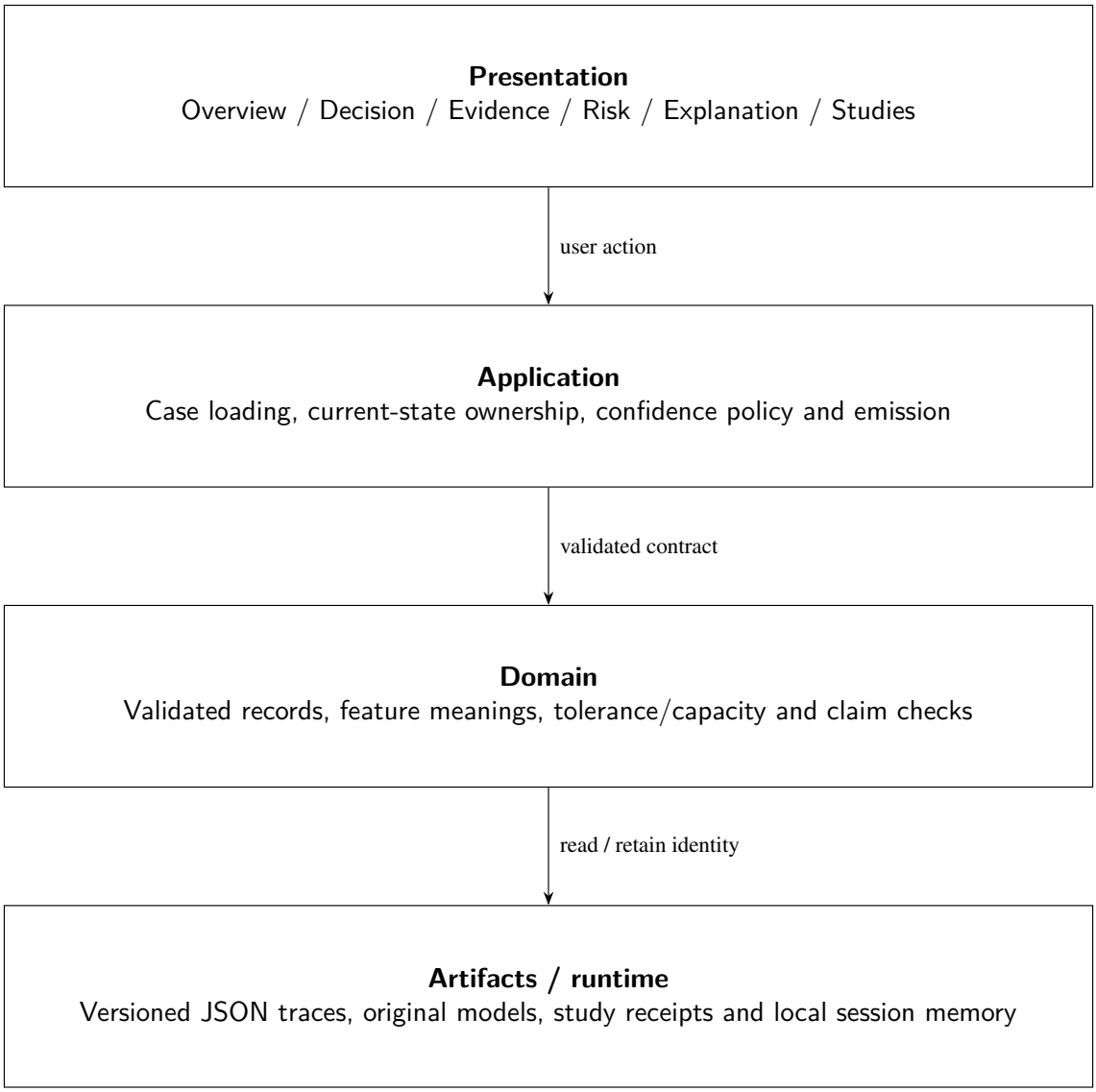
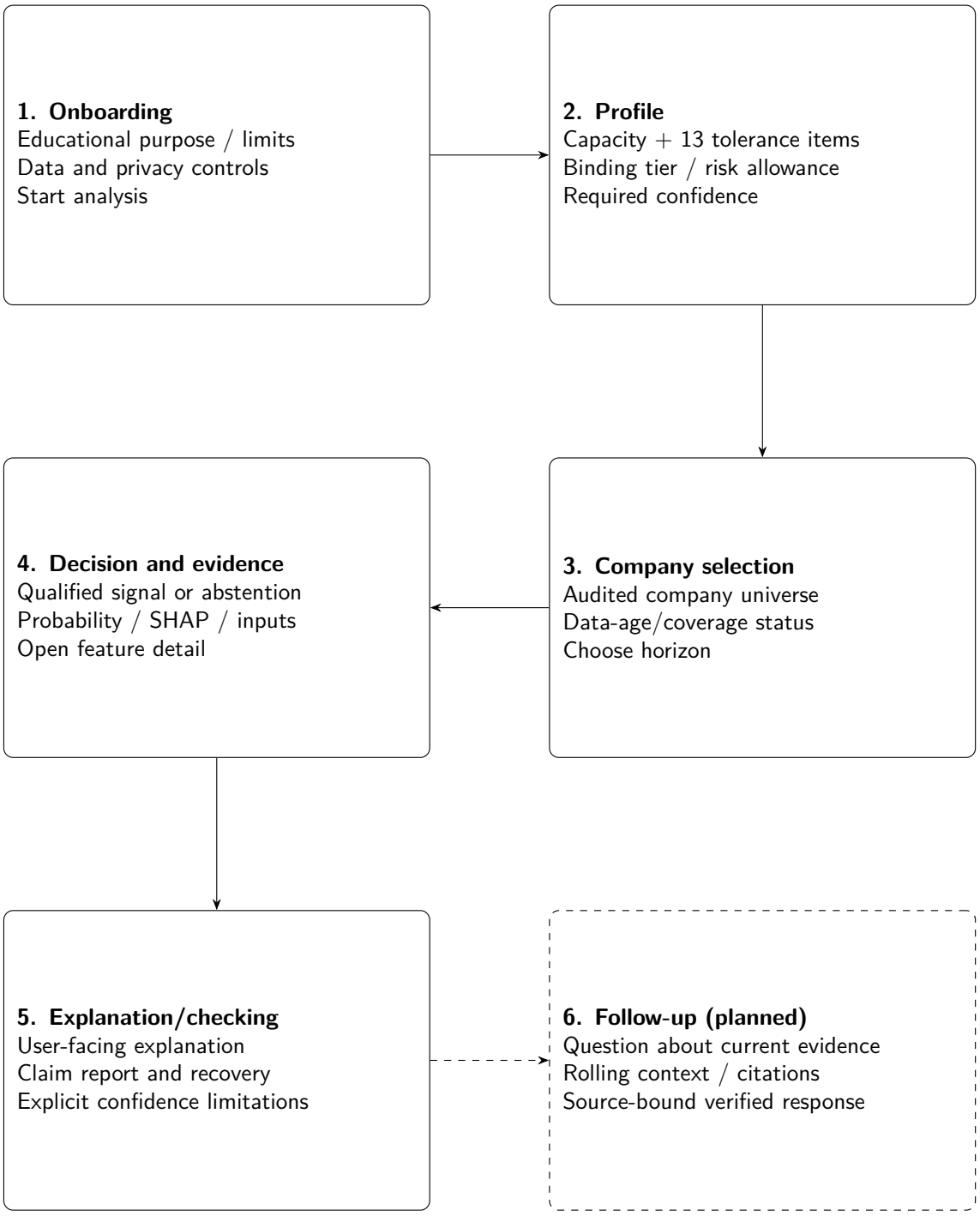


Figure E.4: Current presentation, application, domain and storage layers.



Wireframes describe the product journey. Dashed follow-up and multi-company/service integrations are future scope; actual screenshots are shown separately.

Figure E.5: Planned full-product interfaces; not completed software.

Appendix F

Meeting and Progress Evidence

No formal meeting record was supplied. The form below remains blank.

Table F.1: Meeting log awaiting actual records.

Date	Attendees	Discussion / action	Confirmation

Bibliography

[1] Wealthfront, "How did you create the questionnaire to determine my risk score?" https://support.wealthfront.com/hc/en-us/articles/209353586-How-did-you-create-the-questionnaire-to-determine-my-risk-score, accessed 14 October 2026.

[2] PortfolioPilot, "Personalized recommendations," https://portfoliopilot.com/help/tutorials/tutorial-personalized-recommendations, accessed 14 October 2026.

[3] Danelfin, "How should i use the technical, fundamental, and sentiment subscores?" https://help.danelfin.com/en/articles/what-are-the-technical-fundamental-and-sentiment-subscores, accessed 14 October 2026.

[4] Magnifi, "Ai investing app and financial advisor," https://magnifi.com/, accessed 14 October 2026.

[5] Bloomberg, "News: Bloomberg professional services," https://professional.bloomberg.com/products/bloomberg-terminal/news/, accessed 14 October 2026.

[6] LSEG, "Lseg workspace," https://www.lseg.com/en/data-analytics/products/workspace, accessed 14 October 2026.

[7] Stocktwits, "About stocktwits," https://stocktwits.com/about, accessed 14 October 2026.

[8] TradingView, "Tradingview features," https://www.tradingview.com/features/, accessed 14 October 2026.

[9] L. Ruan and H. Jiang, "Stock price prediction using finbert-enhanced sentiment with shap explainability and differential privacy," *Mathematics*, vol. 13, no. 17, p. 2747, 2025, article number; publisher metadata checked 14 October 2026.

[10] X. Li, Y. Zeng, X. Xing, J. Xu, and X. Xu, "Profit mirage: Revisiting information leakage in llm-based financial agents," https://arxiv.org/abs/2510.07920, 2025, arXiv preprint; primary metadata checked 14 October 2026.

[11] F. Zhang, Z. Li, S. Peng, and Y. Chen, "When alpha disappears: A one-switch benchmark for decision-time leakage in financial backtests," https://arxiv.org/ab

s/2605.23959, 2026, preprint, not peer reviewed; primary metadata checked 14 October 2026.

[12] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.

[13] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[14] S. M. Lundberg, G. G. Erion, and S.-I. Lee, "Consistent individualized feature attribution for tree ensembles," https://arxiv.org/abs/1802.03888, 2018, primary abstract accessed 11 October 2026.

[15] M. L. de Prado, *Advances in Financial Machine Learning*. Wiley, 2018, publisher catalogue checked 14 October 2026.

[16] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "Qlora: Efficient finetuning of quantized llms," https://arxiv.org/abs/2305.14314, 2023.

[17] M. Abdin, J. Aneja, H. Awadalla *et al.*, "Phi-3 technical report: A highly capable language model locally on your phone," https://arxiv.org/abs/2404.14219, 2024.

[18] U.S. Securities and Exchange Commission, "Types of orders," https://www.investor.gov/introduction-investing/investing-basics/how-stock-markets-work/types-orders, accessed 14 October 2026.

[19] ——, "Executing an order," https://www.investor.gov/introduction-investing/investing-basics/how-stock-markets-work/executing-order, accessed 14 October 2026.

[20] ——, "Market participants," https://www.investor.gov/introduction-investing/investing-basics/how-stock-markets-work/market-participants, accessed 14 October 2026.

[21] New York Stock Exchange, "Holidays and trading hours," https://www.nyse.com/trade/hours-calendars, accessed 11 October 2026; historical alignment still uses the actual exchange calendar.

[22] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.

[23] Fidelity Investments, "Relative strength index," https://www.fidelity.com/learning-center/trading-investing/technical-analysis/technical-indicator-guide/rsi, accessed 11 October 2026.

[24] ——, "Moving average convergence divergence," https://www.fidelity.com/learning-center/trading-investing/technical-analysis/technical-indicator-guide/macd, accessed 11 October 2026.

[25] ——, "On balance volume," https://www.fidelity.com/learning-center/trading-investing/technical-analysis/technical-indicator-guide/obv, accessed 11 October 2026.

[26] D. Araci, "Finbert: Financial sentiment analysis with pre-trained language models," https://arxiv.org/abs/1908.10063, 2019, primary abstract accessed 11 October 2026.

[27] Prosus AI, "Prosusai/finbert model card," https://huggingface.co/ProsusAI/finbert, accessed 11 October 2026.

[28] SEC, NASAA, and FINRA, "Artificial intelligence and investment fraud: Investor alert," https://www.investor.gov/introduction-investing/general-resources/news-alerts/alerts-bulletins/investor-alerts/artificial-intelligence-fraud, 2024, accessed 11 October 2026.

[29] NVIDIA, "Nvidia 2021 stock split: Frequently asked questions," https://investor.nvidia.com/files/doc_downloads/doc_faq/06/21/NVIDIA-2021-Stock-Split-FAQ.pdf, 2021, accessed 11 October 2026.

[30] ——, "Nvidia 2024 stock split faq," https://investor.nvidia.com/files/doc_downloads/2024/06/nvidia-2024-stock-split_faq_investors.pdf, 2024.

[31] ElKulako, "Cryptobert model card and label configuration," https://huggingface.co/ElKulako/cryptobert, original paired equity-transfer study executed; current MIT publisher card checked 14 October 2026.

[32] Cardiff NLP, "Twitter-roberta sentiment model card," https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest, accessed 11 October 2026; original pinned revision executed 12 October 2026; Reference-label agreement remains exploratory.

[33] Microsoft, "Phi-3-mini-4k-instruct model card," https://huggingface.co/microsoft/Phi-3-mini-4k-instruct, mIT publisher card; candidate family, project checkpoint not selected; accessed 14 October 2026.

[34] Mistral AI, "Mistral-7b-instruct-v0.3 model card," https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3, apache 2.0 publisher candidate, not an executed project fallback; accessed 14 October 2026.

[35] J. E. Grable and R. H. Lytton, "Financial risk tolerance revisited: the development of a risk assessment instrument," *Financial Services Review*, vol. 8, no. 3, pp. 163–181, 1999.

[36] S. Kuzniak, A. Rabbani, W. Heo, J. Ruiz-Menjivar, and J. E. Grable, "The grable and lytton risk-tolerance scale: A 15-year retrospective," *Financial Services Review*, vol. 24, pp. 177–192, 2015.